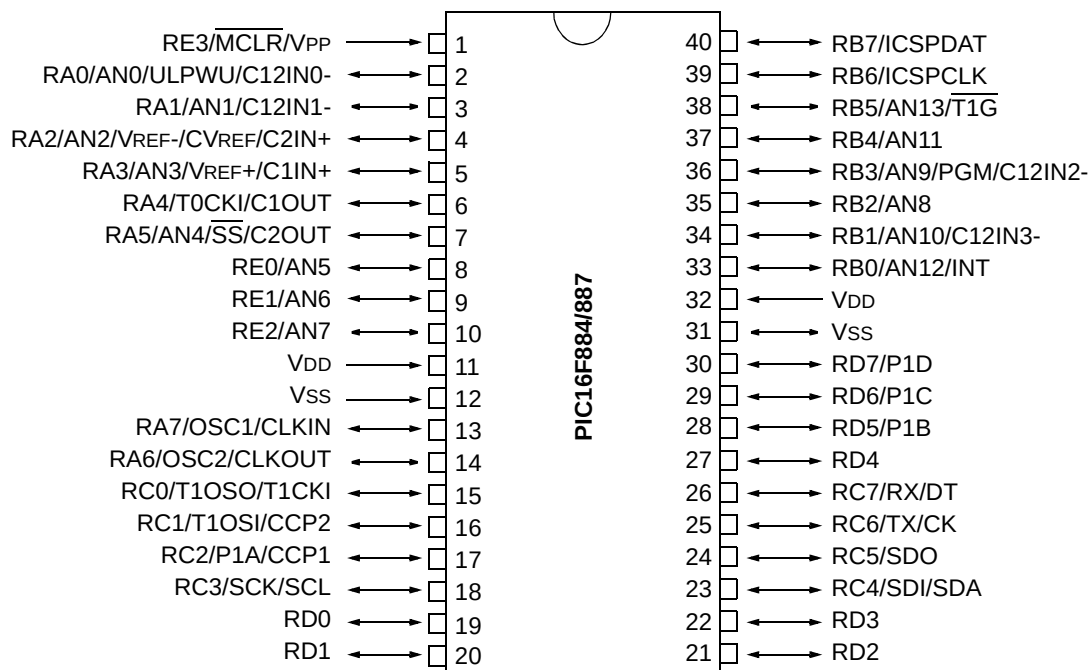


PIC16F882/883/884/886/887

Pin Diagrams – PIC16F884/887, 40-Pin PDIP

40-pin PDIP



PIC16F882/883/884/886/887

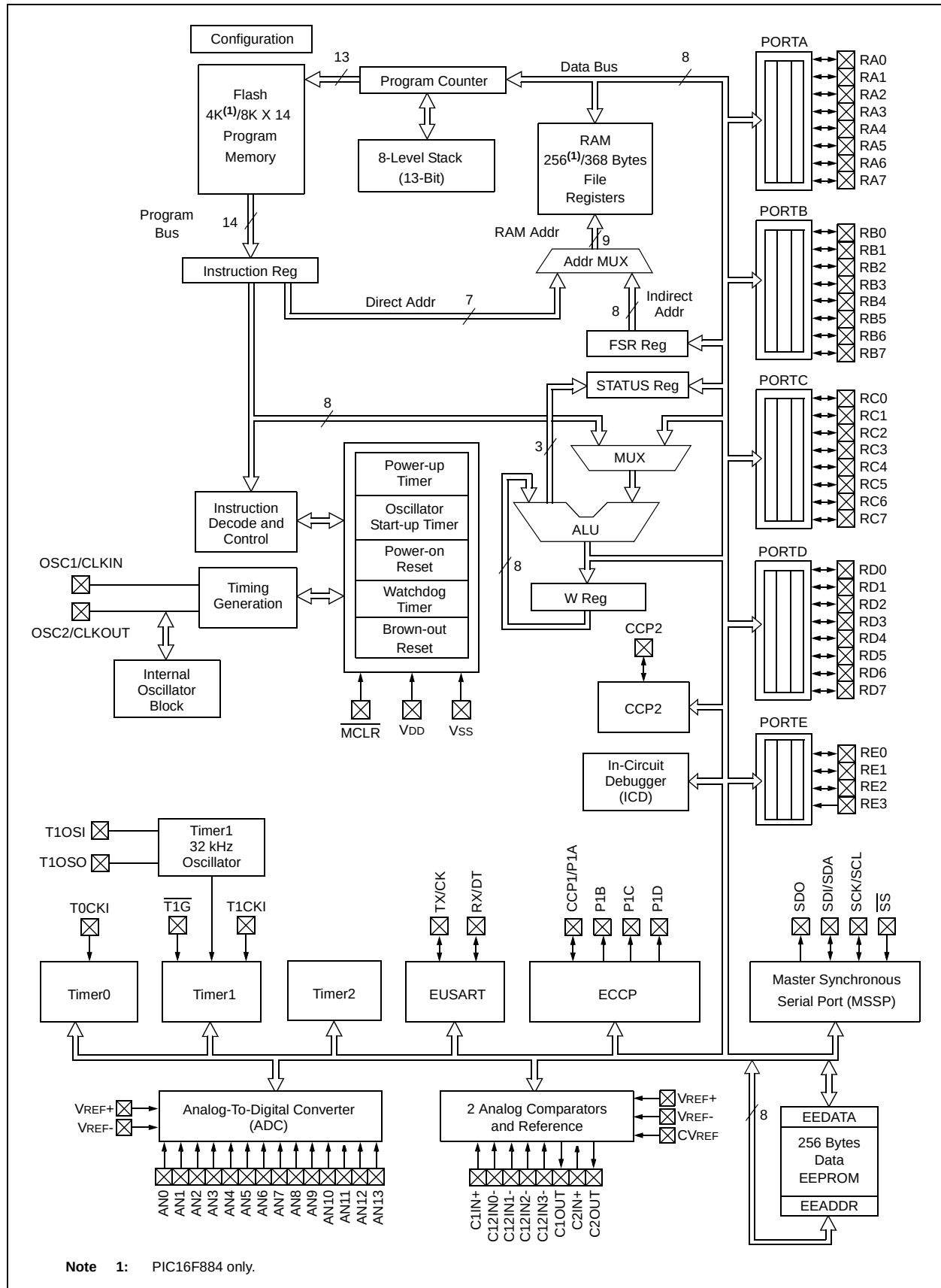
TABLE 3: PIC16F884/887 40-PIN SUMMARY (PDIP)

I/O	Pin	Analog	Comparators	Timers	ECCP	EUSART	MSSP	Interrupt	Pull-up	Basic
RA0	2	AN0/ULPWU	C12IN0-	—	—	—	—	—	—	—
RA1	3	AN1	C12IN1-	—	—	—	—	—	—	—
RA2	4	AN2	C2IN+	—	—	—	—	—	—	VREF-/CVREF
RA3	5	AN3	C1IN+	—	—	—	—	—	—	VREF+
RA4	6	—	C1OUT	T0CKI	—	—	—	—	—	—
RA5	7	AN4	C2OUT	—	—	—	SS	—	—	—
RA6	14	—	—	—	—	—	—	—	—	OSC2/CLKOUT
RA7	13	—	—	—	—	—	—	—	—	OSC1/CLKIN
RB0	33	AN12	—	—	—	—	—	IOC/INT	Y	—
RB1	34	AN10	C12IN3-	—	—	—	—	IOC	Y	—
RB2	35	AN8	—	—	—	—	—	IOC	Y	—
RB3	36	AN9	C12IN2-	—	—	—	—	IOC	Y	PGM
RB4	37	AN11	—	—	—	—	—	IOC	Y	—
RB5	38	AN13	—	T1G	—	—	—	IOC	Y	—
RB6	39	—	—	—	—	—	—	IOC	Y	ICSPCLK
RB7	40	—	—	—	—	—	—	IOC	Y	ICSPDAT
RC0	15	—	—	T1OSO/T1CKI	—	—	—	—	—	—
RC1	16	—	—	T1OSI	CCP2	—	—	—	—	—
RC2	17	—	—	—	CCP1/P1A	—	—	—	—	—
RC3	18	—	—	—	—	—	SCK/SCL	—	—	—
RC4	23	—	—	—	—	—	SDI/SDA	—	—	—
RC5	24	—	—	—	—	—	SDO	—	—	—
RC6	25	—	—	—	—	TX/CK	—	—	—	—
RC7	26	—	—	—	—	RX/DT	—	—	—	—
RD0	19	—	—	—	—	—	—	—	—	—
RD1	20	—	—	—	—	—	—	—	—	—
RD2	21	—	—	—	—	—	—	—	—	—
RD3	22	—	—	—	—	—	—	—	—	—
RD4	27	—	—	—	—	—	—	—	—	—
RD5	28	—	—	—	P1B	—	—	—	—	—
RD6	29	—	—	—	P1C	—	—	—	—	—
RD7	30	—	—	—	P1D	—	—	—	—	—
RE0	8	AN5	—	—	—	—	—	—	—	—
RE1	9	AN6	—	—	—	—	—	—	—	—
RE2	10	AN7	—	—	—	—	—	—	—	—
RE3	1	—	—	—	—	—	—	—	Y ⁽¹⁾	MCLR/VPP
—	11	—	—	—	—	—	—	—	—	VDD
—	32	—	—	—	—	—	—	—	—	VDD
—	12	—	—	—	—	—	—	—	—	VSS
—	31	—	—	—	—	—	—	—	—	VSS

Note 1: Pull-up activated only with external MCLR configuration.

PIC16F882/883/884/886/887

FIGURE 1-2: PIC16F884/PIC16F887 BLOCK DIAGRAM



PIC16F882/883/884/886/887

TABLE 1-2: PIC16F884/887 PINOUT DESCRIPTION

Name	Function	Input Type	Output Type	Description
RA0/AN0/ULPWU/C12IN0-	RA0	TTL	CMOS	General purpose I/O.
	AN0	AN	—	A/D Channel 0 input.
	ULPWU	AN	—	Ultra Low-Power Wake-up input.
	C12IN0-	AN	—	Comparator C1 or C2 negative input.
RA1/AN1/C12IN1-	RA1	TTL	CMOS	General purpose I/O.
	AN1	AN	—	A/D Channel 1 input.
	C12IN1-	AN	—	Comparator C1 or C2 negative input.
RA2/AN2/VREF-/CVREF/C2IN+	RA2	TTL	CMOS	General purpose I/O.
	AN2	AN	—	A/D Channel 2.
	VREF-	AN	—	A/D Negative Voltage Reference input.
	CVREF	—	AN	Comparator Voltage Reference output.
	C2IN+	AN	—	Comparator C2 positive input.
RA3/AN3/VREF+/C1IN+	RA3	TTL	CMOS	General purpose I/O.
	AN3	AN	—	A/D Channel 3.
	VREF+	AN	—	A/D Positive Voltage Reference input.
	C1IN+	AN	—	Comparator C1 positive input.
RA4/T0CKI/C1OUT	RA4	TTL	CMOS	General purpose I/O.
	T0CKI	ST	—	Timer0 clock input.
	C1OUT	—	CMOS	Comparator C1 output.
RA5/AN4/ $\overline{SS}$ /C2OUT	RA5	TTL	CMOS	General purpose I/O.
	AN4	AN	—	A/D Channel 4.
	$\overline{SS}$	ST	—	Slave Select input.
	C2OUT	—	CMOS	Comparator C2 output.
RA6/OSC2/CLKOUT	RA6	TTL	CMOS	General purpose I/O.
	OSC2	—	XTAL	Crystal/Resonator.
	CLKOUT	—	CMOS	Fosc/4 output.
RA7/OSC1/CLKIN	RA7	TTL	CMOS	General purpose I/O.
	OSC1	XTAL	—	Crystal/Resonator.
	CLKIN	ST	—	External clock input/RC oscillator connection.
RB0/AN12/INT	RB0	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	AN12	AN	—	A/D Channel 12.
	INT	ST	—	External interrupt.
RB1/AN10/C12IN3-	RB1	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	AN10	AN	—	A/D Channel 10.
	C12IN3-	AN	—	Comparator C1 or C2 negative input.
RB2/AN8	RB2	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	AN8	AN	—	A/D Channel 8.
RB3/AN9/PGM/C12IN2-	RB3	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	AN9	AN	—	A/D Channel 9.
	PGM	ST	—	Low-voltage ICSP™ Programming enable pin.
	C12IN2-	AN	—	Comparator C1 or C2 negative input.

Legend: AN = Analog input or output CMOS = CMOS compatible input or output OD = Open Drain
TTL = TTL compatible input ST = Schmitt Trigger input with CMOS levels
HV = High Voltage XTAL = Crystal

PIC16F882/883/884/886/887

TABLE 1-2: PIC16F884/887 PINOUT DESCRIPTION (CONTINUED)

Name	Function	Input Type	Output Type	Description
RB4/AN11	RB4	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	AN11	AN	—	A/D Channel 11.
RB5/AN13/ $\overline{T1G}$	RB5	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	AN13	AN	—	A/D Channel 13.
	$\overline{T1G}$	ST	—	Timer1 Gate input.
RB6/ICSPCLK	RB6	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	ICSPCLK	ST	—	Serial Programming Clock.
RB7/ICSPDAT	RB7	TTL	CMOS	General purpose I/O. Individually controlled interrupt-on-change. Individually enabled pull-up.
	ICSPDAT	ST	TTL	ICSP™ Data I/O.
RC0/T1OSO/T1CKI	RC0	ST	CMOS	General purpose I/O.
	T1OSO	—	XTAL	Timer1 oscillator output.
	T1CKI	ST	—	Timer1 clock input.
RC1/T1OSI/CCP2	RC1	ST	CMOS	General purpose I/O.
	T1OSI	XTAL	—	Timer1 oscillator input.
	CCP2	ST	CMOS	Capture/Compare/PWM2.
RC2/P1A/CCP1	RC2	ST	CMOS	General purpose I/O.
	P1A	ST	CMOS	PWM output.
	CCP1	—	CMOS	Capture/Compare/PWM1.
RC3/SCK/SCL	RC3	ST	CMOS	General purpose I/O.
	SCK	ST	CMOS	SPI clock.
	SCL	ST	OD	I ² C™ clock.
RC4/SDI/SDA	RC4	ST	CMOS	General purpose I/O.
	SDI	ST	—	SPI data input.
	SDA	ST	OD	I ² C data input/output.
RC5/SDO	RC5	ST	CMOS	General purpose I/O.
	SDO	—	CMOS	SPI data output.
RC6/TX/CK	RC6	ST	CMOS	General purpose I/O.
	TX	—	CMOS	EUSART asynchronous transmit.
	CK	ST	CMOS	EUSART synchronous clock.
RC7/RX/DT	RC7	ST	CMOS	General purpose I/O.
	RX	ST	—	EUSART asynchronous input.
	DT	ST	CMOS	EUSART synchronous data.
RD0	RD0	TTL	CMOS	General purpose I/O.
RD1	RD1	TTL	CMOS	General purpose I/O.
RD2	RD2	TTL	CMOS	General purpose I/O.
RD3	RD3	TTL	CMOS	General purpose I/O.
RD4	RD4	TTL	CMOS	General purpose I/O.
RD5/P1B	RD5	TTL	CMOS	General purpose I/O.
	P1B	—	CMOS	PWM output.
RD6/P1C	RD6	TTL	CMOS	General purpose I/O.
	P1C	—	CMOS	PWM output.

Legend: AN = Analog input or output CMOS = CMOS compatible input or output OD = Open Drain
TTL = TTL compatible input ST = Schmitt Trigger input with CMOS levels
HV = High Voltage XTAL = Crystal

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TABLE 1-2: PIC16F884/887 PINOUT DESCRIPTION (CONTINUED)

Name	Function	Input Type	Output Type	Description
RD7/P1D	RD7	TTL	CMOS	General purpose I/O.
	P1D	AN	—	PWM output.
RE0/AN5	RE0	TTL	CMOS	General purpose I/O.
	AN5	AN	—	A/D Channel 5.
RE1/AN6	RE1	TTL	CMOS	General purpose I/O.
	AN6	AN	—	A/D Channel 6.
RE2/AN7	RE2	TTL	CMOS	General purpose I/O.
	AN7	AN	—	A/D Channel 7.
RE3/MCLR/VPP	RE3	TTL	—	General purpose input.
	MCLR	ST	—	Master Clear with internal pull-up.
	VPP	HV	—	Programming voltage.
VSS	VSS	Power	—	Ground reference.
VDD	VDD	Power	—	Positive supply.

Legend: AN = Analog input or output CMOS = CMOS compatible input or output OD = Open Drain
TTL = TTL compatible input ST = Schmitt Trigger input with CMOS levels
HV = High Voltage XTAL = Crystal

2.0 MEMORY ORGANIZATION

2.1 Program Memory Organization

The PIC16F882/883/884/886/887 has a 13-bit program counter capable of addressing a 2K x 14 (0000h-07FFh) for the PIC16F882, 4K x 14 (0000h-0FFFh) for the PIC16F883/PIC16F884, and 8K x 14 (0000h-1FFFh) for the PIC16F886/PIC16F887 program memory space. Accessing a location above these boundaries will cause a wrap-around within the first 8K x 14 space. The Reset vector is at 0000h and the interrupt vector is at 0004h (see Figures 2-2 and 2-3).

FIGURE 2-1: PROGRAM MEMORY MAP AND STACK FOR THE PIC16F882

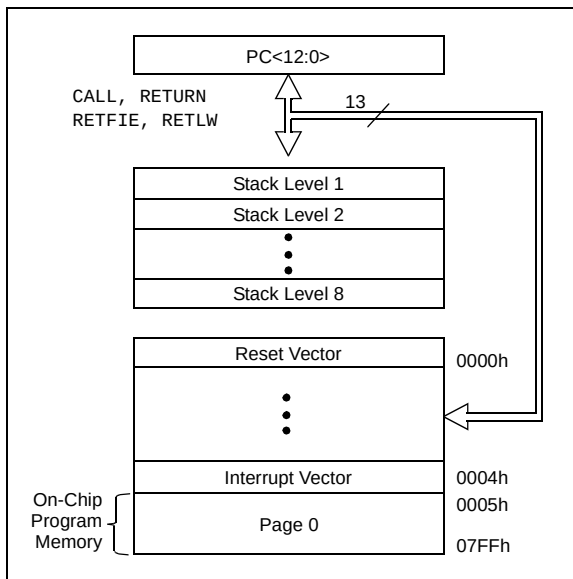


FIGURE 2-2: PROGRAM MEMORY MAP AND STACK FOR THE PIC16F883/PIC16F884

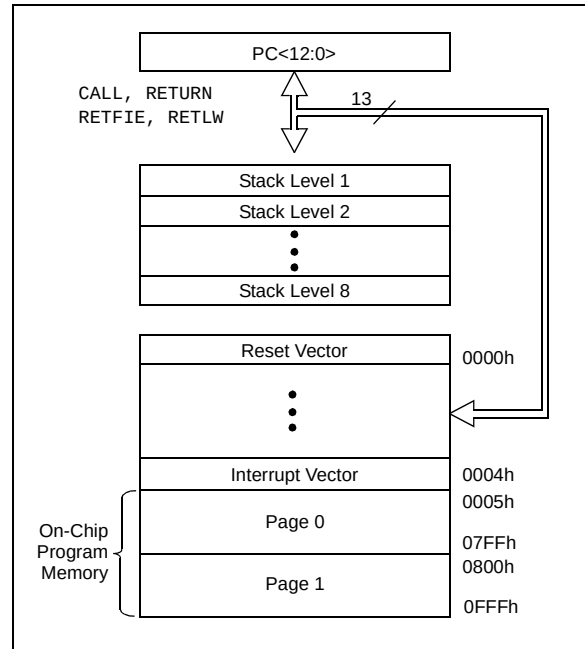
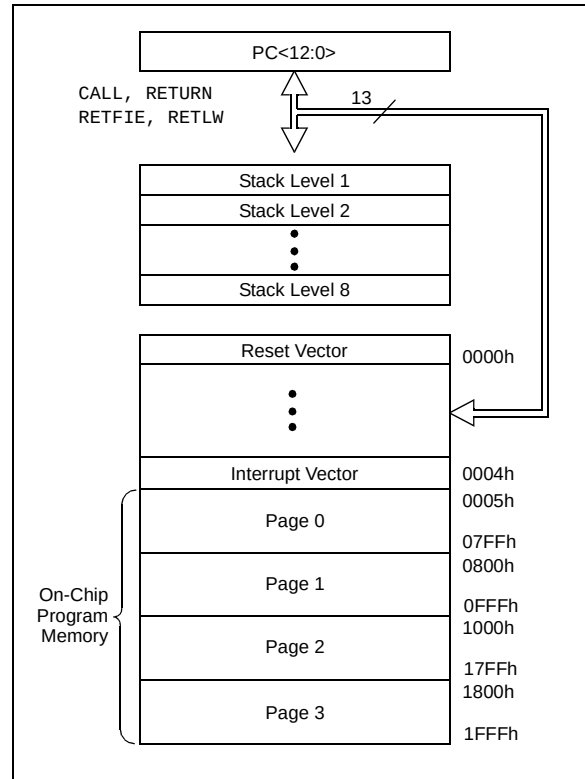


FIGURE 2-3: PROGRAM MEMORY MAP AND STACK FOR THE PIC16F886/PIC16F887



2.2 Data Memory Organization

The data memory (see Figures 2-2 and 2-3) is partitioned into four banks which contain the General Purpose Registers (GPR) and the Special Function Registers (SFR). The Special Function Registers are located in the first 32 locations of each bank. The General Purpose Registers, implemented as static RAM, are located in the last 96 locations of each Bank. Register locations F0h-FFh in Bank 1, 170h-17Fh in Bank 2 and 1F0h-1FFh in Bank 3, point to addresses 70h-7Fh in Bank 0. The actual number of General Purpose Registers (GPR) implemented in each Bank depends on the device. Details are shown in Figures 2-5 and 2-6. All other RAM is unimplemented and returns '0' when read. RP<1:0> of the STATUS register are the bank select bits:

RP1 RP0

0	0	→Bank 0 is selected
0	1	→Bank 1 is selected
1	0	→Bank 2 is selected
1	1	→Bank 3 is selected

2.2.1 GENERAL PURPOSE REGISTER FILE

The register file is organized as 128 x 8 in the PIC16F882, 256 x 8 in the PIC16F883/PIC16F884, and 368 x 8 in the PIC16F886/PIC16F887. Each register is accessed, either directly or indirectly, through the File Select Register (FSR) (see **Section 2.4 “Indirect Addressing, INDF and FSR Registers”**).

2.2.2 SPECIAL FUNCTION REGISTERS

The Special Function Registers are registers used by the CPU and peripheral functions for controlling the desired operation of the device (see Table 2-1). These registers are static RAM.

The special registers can be classified into two sets: core and peripheral. The Special Function Registers associated with the “core” are described in this section. Those related to the operation of the peripheral features are described in the section of that peripheral feature.

PIC16F882/883/884/886/887

FIGURE 2-6: PIC16F886/PIC16F887 SPECIAL FUNCTION REGISTERS

File		File		File		File			
Address		Address		Address		Address			
Indirect addr. ⁽¹⁾	00h	Indirect addr. ⁽¹⁾	80h	Indirect addr. ⁽¹⁾	100h	Indirect addr. ⁽¹⁾	180h		
TMR0	01h	OPTION_REG	81h	TMR0	101h	OPTION_REG	181h		
PCL	02h	PCL	82h	PCL	102h	PCL	182h		
STATUS	03h	STATUS	83h	STATUS	103h	STATUS	183h		
FSR	04h	FSR	84h	FSR	104h	FSR	184h		
PORTA	05h	TRISA	85h	WDTCON	105h	SRCON	185h		
PORTB	06h	TRISB	86h	PORTB	106h	TRISB	186h		
PORTC	07h	TRISC	87h	CM1CON0	107h	BAUDCTL	187h		
PORTD ⁽²⁾	08h	TRISD ⁽²⁾	88h	CM2CON0	108h	ANSEL	188h		
PORTE	09h	TRISE	89h	CM2CON1	109h	ANSELH	189h		
PCLATH	0Ah	PCLATH	8Ah	PCLATH	10Ah	PCLATH	18Ah		
INTCON	0Bh	INTCON	8Bh	INTCON	10Bh	INTCON	18Bh		
PIR1	0Ch	PIE1	8Ch	EEDAT	10Ch	EECON1	18Ch		
PIR2	0Dh	PIE2	8Dh	EEADR	10Dh	EECON2 ⁽¹⁾	18Dh		
TMR1L	0Eh	PCON	8Eh	EEDATH	10Eh	Reserved	18Eh		
TMR1H	0Fh	OSCCON	8Fh	EEADRH	10Fh	Reserved	18Fh		
T1CON	10h	OSCTUNE	90h	General Purpose Registers 16 Bytes	110h	General Purpose Registers 16 Bytes	190h		
TMR2	11h	SSPCON2	91h		111h		191h		
T2CON	12h	PR2	92h		112h		192h		
SSPBUF	13h	SSPADD	93h		113h		193h		
SSPCON	14h	SSPSTAT	94h		114h		194h		
CCPR1L	15h	WPUB	95h		115h		195h		
CCPR1H	16h	IOCB	96h		116h		196h		
CCP1CON	17h	VRCON	97h		117h		197h		
RCSTA	18h	TXSTA	98h		118h		198h		
TXREG	19h	SPBRG	99h		119h		199h		
RCREG	1Ah	SPBRGH	9Ah		11Ah		19Ah		
CCPR2L	1Bh	PWM1CON	9Bh		11Bh		19Bh		
CCPR2H	1Ch	ECCPAS	9Ch		11Ch		19Ch		
CCP2CON	1Dh	PSTRCON	9Dh		11Dh		19Dh		
ADRESH	1Eh	ADRESL	9Eh		11Eh		19Eh		
ADCON0	1Fh	ADCON1	9Fh		11Fh		19Fh		
General Purpose Registers 96 Bytes	20h	General Purpose Registers 80 Bytes	A0h	General Purpose Registers 80 Bytes	120h	General Purpose Registers 80 Bytes	1A0h		
	3Fh		EFh		16Fh		1EFh		
	40h								
	6Fh								
	70h							accesses 70h-7Fh	F0h
	7Fh		FFh		17Fh		1FFh		
Bank 0		Bank 1		Bank 2		Bank 3			

Unimplemented data memory locations, read as '0'.

Note 1: Not a physical register.
2: PIC16F887 only.

PIC16F882/883/884/886/887

TABLE 2-1: PIC16F882/883/884/886/887 SPECIAL FUNCTION REGISTERS SUMMARY BANK 0

Addr	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Page
Bank 0											
00h	INDF	Addressing this location uses contents of FSR to address data memory (not a physical register)								xxxx xxxx	37,217
01h	TMR0	Timer0 Module Register								xxxx xxxx	73,217
02h	PCL	Program Counter's (PC) Least Significant Byte								0000 0000	37,217
03h	STATUS	IRP	RP1	RP0	$\overline{TO}$	$\overline{PD}$	Z	DC	C	0001 1xxx	29,217
04h	FSR	Indirect Data Memory Address Pointer								xxxx xxxx	37,217
05h	PORTA ⁽³⁾	RA7	RA6	RA5	RA4	RA3	RA2	RA1	RA0	xxxx xxxx	39,217
06h	PORTB ⁽³⁾	RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0	xxxx xxxx	48,217
07h	PORTC ⁽³⁾	RC7	RC6	RC5	RC4	RC3	RC2	RC1	RC0	xxxx xxxx	53,217
08h	PORTD ^(3,4)	RD7	RD6	RD5	RD4	RD3	RD2	RD1	RD0	xxxx xxxx	57,217
09h	PORTE ⁽³⁾	—	—	—	—	RE3	RE2 ⁽⁴⁾	RE1 ⁽⁴⁾	RE0 ⁽⁴⁾	--- xxxx	59,217
0Ah	PCLATH	—	—	—	Write Buffer for upper 5 bits of Program Counter					---0 0000	37,217
0Bh	INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBFIF ⁽¹⁾	0000 000x	31,217
0Ch	PIR1	—	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	-000 0000	34,217
0Dh	PIR2	OSFIF	C2IF	C1IF	EEIF	BCLIF	ULPWUIF	—	CCP2IF	0000 00-0	35,217
0Eh	TMR1L	Holding Register for the Least Significant Byte of the 16-bit TMR1 Register								xxxx xxxx	76,217
0Fh	TMR1H	Holding Register for the Most Significant Byte of the 16-bit TMR1 Register								xxxx xxxx	76,217
10h	T1CON	T1GINV	TMR1GE	T1CKPS1	T1CKPS0	T1OSCEN	$\overline{T1SYNC}$	TMR1CS	TMR1ON	0000 0000	79,217
11h	TMR2	Timer2 Module Register								0000 0000	81,217
12h	T2CON	—	TOUTPS3	TOUTPS2	TOUTPS1	TOUTPS0	TMR2ON	T2CKPS1	T2CKPS0	-000 0000	82,217
13h	SSPBUF	Synchronous Serial Port Receive Buffer/Transmit Register								xxxx xxxx	183,217
14h	SSPCON ⁽²⁾	WCOL	SSPOV	SSPEN	CKP	SSPM3	SSPM2	SSPM1	SSPM0	0000 0000	181,217
15h	CCPR1L	Capture/Compare/PWM Register 1 Low Byte (LSB)								xxxx xxxx	126,217
16h	CCPR1H	Capture/Compare/PWM Register 1 High Byte (MSB)								xxxx xxxx	126,217
17h	CCP1CON	P1M1	P1M0	DC1B1	DC1B0	CCP1M3	CCP1M2	CCP1M1	CCP1M0	0000 0000	124,217
18h	RCSTA	SPEN	RX9	SREN	CREN	ADDEN	FERR	OERR	RX9D	0000 000x	161,217
19h	TXREG	EUSART Transmit Data Register								0000 0000	153,217
1Ah	RCREG	EUSART Receive Data Register								0000 0000	158,217
1Bh	CCPR2L	Capture/Compare/PWM Register 2 Low Byte (LSB)								xxxx xxxx	126,217
1Ch	CCPR2H	Capture/Compare/PWM Register 2 High Byte (MSB)								xxxx xxxx	126,218
1Dh	CCP2CON	—	—	DC2B1	DC2B0	CCP2M3	CCP2M2	CCP2M1	CCP2M0	--00 0000	125,218
1Eh	ADRESH	A/D Result Register High Byte								xxxx xxxx	99,218
1Fh	ADCON0	ADCS1	ADCS0	CHS3	CHS2	CHS1	CHS0	$\overline{GO/DONE}$	ADON	0000 0000	104,218

- Legend:** — = Unimplemented locations read as '0', u = unchanged, x = unknown, q = value depends on condition, shaded = unimplemented
- Note** 1: MCLR and WDT Reset do not affect the previous value data latch. The RBIF bit will be cleared upon Reset but will set again if the mismatch exists.
- 2: When SSPCON register bits SSPM<3:0> = 1001, any reads or writes to the SSPADD SFR address are accessed through the SSPMSK register. See Registers • and 13-4 for more detail.
- 3: Port pins with analog functions controlled by the ANSEL and ANSELH registers will read '0' immediately after a Reset even though the data latches are either undefined (POR) or unchanged (other Resets).
- 4: PIC16F884/PIC16F887 only.

PIC16F882/883/884/886/887

TABLE 2-2: PIC16F882/883/884/886/887 SPECIAL FUNCTION REGISTERS SUMMARY BANK 1

Addr	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Page
Bank 1											
80h	INDF	Addressing this location uses contents of FSR to address data memory (not a physical register)								xxxx xxxx	37,217
81h	OPTION_REG	RBPUR	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	30,218
82h	PCL	Program Counter's (PC) Least Significant Byte								0000 0000	37,217
83h	STATUS	IRP	RP1	RP0	T0	PD	Z	DC	C	0001 1xxx	29,217
84h	FSR	Indirect Data Memory Address Pointer								xxxx xxxx	37,217
85h	TRISA	TRISA7	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0	1111 1111	39,218
86h	TRISB	TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0	1111 1111	48,218
87h	TRISC	TRISC7	TRISC6	TRISC5	TRISC4	TRISC3	TRISC2	TRISC1	TRISC0	1111 1111	53,218
88h	TRISD ⁽³⁾	TRISD7	TRISD6	TRISD5	TRISD4	TRISD3	TRISD2	TRISD1	TRISD0	1111 1111	57,218
89h	TRISE	—	—	—	—	TRISE3	TRISE2 ⁽³⁾	TRISE1 ⁽³⁾	TRISE0 ⁽³⁾	---- 1111	59,218
8Ah	PCLATH	—	—	—	Write Buffer for the upper 5 bits of the Program Counter					---0 0000	37,217
8Bh	INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF ⁽¹⁾	0000 000x	31,217
8Ch	PIE1	—	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	-000 0000	32,218
8Dh	PIE2	OSFIE	C2IE	C1IE	EEIE	BCLIE	ULPWUIE	—	CCP2IE	0000 00-0	33,218
8Eh	PCON	—	—	ULPWUE	SBOREN	—	—	POR	BOR	--01 --qq	36,218
8Fh	OSCCON	—	IRCF2	IRCF1	IRCF0	OSTS	HTS	LTS	SCS	-110 q000	62,218
90h	OSCTUNE	—	—	—	TUN4	TUN3	TUN2	TUN1	TUN0	---0 0000	66,218
91h	SSPCON2	GCEN	ACKSTAT	ACKDT	ACKEN	RCEN	PEN	RSEN	SEN	0000 0000	181,218
92h	PR2	Timer2 Period Register								1111 1111	81,218
93h	SSPADD ⁽²⁾	Synchronous Serial Port (I ² C mode) Address Register								0000 0000	189,218
93h	SSPMASK ⁽²⁾	MSK7	MSK6	MSK5	MSK4	MSK3	MSK2	MSK1	MSK0	1111 1111	189,218
94h	SSPSTAT	SMP	CKE	D/A	P	S	R/W	UA	BF	0000 0000	189,218
95h	WPUB	WPUB7	WPUB6	WPUB5	WPUB4	WPUB3	WPUB2	WPUB1	WPUB0	1111 1111	49,218
96h	IOCB	IOCB7	IOCB6	IOCB5	IOCB4	IOCB3	IOCB2	IOCB1	IOCB0	0000 0000	49,218
97h	VRCON	VREN	VROE	VRR	VRSS	VR3	VR2	VR1	VR0	0000 0000	97,218
98h	TXSTA	CSRC	TX9	TXEN	SYNC	SENDB	BRGH	TRMT	TX9D	0000 0010	160,218
99h	SPBRG	BRG7	BRG6	BRG5	BRG4	BRG3	BRG2	BRG1	BRG0	0000 0000	163,218
9Ah	SPBRGH	BRG15	BRG14	BRG13	BRG12	BRG11	BRG10	BRG9	BRG8	0000 0000	163,218
9Bh	PWM1CON	PRSEN	PDC6	PDC5	PDC4	PDC3	PDC2	PDC1	PDC0	0000 0000	145,218
9Ch	ECCPAS	ECCPASE	ECCPAS2	ECCPAS1	ECCPAS0	PSSAC1	PSSAC0	PSSBD1	PSSBD0	0000 0000	142,218
9Dh	PSTRCON	—	—	—	STRSYNC	STRD	STRC	STRB	STRA	---0 0001	146,218
9Eh	ADRESL	A/D Result Register Low Byte								xxxx xxxx	99,218
9Fh	ADCON1	ADFM	—	VCFG1	VCFG0	—	—	—	—	0-00 ----	105,218

Legend: — = Unimplemented locations read as '0', u = unchanged, x = unknown, q = value depends on condition, shaded = unimplemented

Note 1: MCLR and WDT Reset do not affect the previous value data latch. The RBIF bit will be cleared upon Reset but will set again if the mismatch exists.

2: Accessible only when SSPCON register bits SSPM<3:0> = 1001.

3: PIC16F884/PIC16F887 only.

PIC16F882/883/884/886/887

TABLE 2-3: PIC16F882/883/884/886/887 SPECIAL FUNCTION REGISTERS SUMMARY BANK 2

Addr	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Page
Bank 2											
100h	INDF	Addressing this location uses contents of FSR to address data memory (not a physical register)								xxxx xxxx	37,217
101h	TMR0	Timer0 Module Register								xxxx xxxx	73,217
102h	PCL	Program Counter's (PC) Least Significant Byte								0000 0000	37,217
103h	STATUS	IRP	RP1	RP0	$\overline{TO}$	$\overline{PD}$	Z	DC	C	0001 1xxx	29,217
104h	FSR	Indirect Data Memory Address Pointer								xxxx xxxx	37,217
105h	WDTCON	—	—	—	WDTPS3	WDTPS2	WDTPS1	WDTPS0	SWDTEN	---0 1000	225,218
106h	PORTB	RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0	xxxx xxxx	48,217
107h	CM1CON0	C1ON	C1OUT	C1OE	C1POL	—	C1R	C1CH1	C1CH0	0000 -000	88,218
108h	CM2CON0	C2ON	C2OUT	C2OE	C2POL	—	C2R	C2CH1	C2CH0	0000 -000	89,218
109h	CM2CON1	MC1OUT	MC2OUT	C1RSEL	C2RSEL	—	—	T1GSS	C2SYNC	0000 --10	91,219
10Ah	PCLATH	—	—	—	Write Buffer for the upper 5 bits of the Program Counter					---0 0000	37,217
10Bh	INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF ⁽¹⁾	0000 000x	31,217
10Ch	EEDAT	EEDAT7	EEDAT6	EEDAT5	EEDAT4	EEDAT3	EEDAT2	EEDAT1	EEDAT0	0000 0000	112,219
10Dh	EEADR	EEADR7	EEADR6	EEADR5	EEADR4	EEADR3	EEADR2	EEADR1	EEADR0	0000 0000	112,219
10Eh	EEDATH	—	—	EEDATH5	EEDATH4	EEDATH3	EEDATH2	EEDATH1	EEDATH0	--00 0000	112,219
10Fh	EEADRH	—	—	—	EEADRH4 ⁽²⁾	EEADRH3	EEADRH2	EEADRH1	EEADRH0	---- 0000	112,219

Legend: — = Unimplemented locations read as '0', u = unchanged, x = unknown, q = value depends on condition, shaded = unimplemented
Note 1: MCLR and WDT Reset does not affect the previous value data latch. The RBIF bit will be cleared upon Reset but will set again if the mismatch exists.
2: PIC16F886/PIC16F887 only.

TABLE 2-4: PIC16F882/883/884/886/887 SPECIAL FUNCTION REGISTERS SUMMARY BANK 3

Addr	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Page
Bank 3											
180h	INDF	Addressing this location uses contents of FSR to address data memory (not a physical register)								xxxx xxxx	37,217
181h	OPTION_REG	RBPV	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	30,218
182h	PCL	Program Counter's (PC) Least Significant Byte								0000 0000	37,217
183h	STATUS	IRP	RP1	RP0	$\overline{TO}$	$\overline{PD}$	Z	DC	C	0001 1xxx	29,217
184h	FSR	Indirect Data Memory Address Pointer								xxxx xxxx	37,217
185h	SRCON	SR1	SR0	C1SEN	C2REN	PULSS	PULSR	—	FVREN	0000 00-0	93,219
186h	TRISB	TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0	1111 1111	48,218
187h	BAUDCTL	ABDOVF	RCIDL	—	SCKP	BRG16	—	WUE	ABDEN	01-0 0-00	162,219
188h	ANSEL	ANS7 ⁽²⁾	ANS6 ⁽²⁾	ANS5 ⁽²⁾	ANS4	ANS3	ANS2	ANS1	ANS0	1111 1111	40,219
189h	ANSELH	—	—	ANS13	ANS12	ANS11	ANS10	ANS9	ANS8	--11 1111	99,219
18Ah	PCLATH	—	—	—	Write Buffer for the upper 5 bits of the Program Counter					---0 0000	37,217
18Bh	INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF ⁽¹⁾	0000 000x	31,217
18Ch	EECON1	EEPGD	—	—	—	WRERR	WREN	WR	RD	x--- x000	113,219
18Dh	EECON2	EEPROM Control Register 2 (not a physical register)								---- ----	111,219

Legend: — = Unimplemented locations read as '0', u = unchanged, x = unknown, q = value depends on condition, shaded = unimplemented
Note 1: MCLR and WDT Reset does not affect the previous value data latch. The RBIF bit will be cleared upon Reset but will set again if the mismatch exists.
2: PIC16F884/PIC16F887 only.

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2.2.2.1 STATUS Register

The STATUS register, shown in Register 2-1, contains:

- the arithmetic status of the ALU
- the Reset status
- the bank select bits for data memory (GPR and SFR)

The STATUS register can be the destination for any instruction, like any other register. If the STATUS register is the destination for an instruction that affects the Z, DC or C bits, then the write to these three bits is disabled. These bits are set or cleared according to the device logic. Furthermore, the $\overline{\text{TO}}$ and $\overline{\text{PD}}$ bits are not writable. Therefore, the result of an instruction with the STATUS register as destination may be different than intended.

For example, CLRF STATUS, will clear the upper three bits and set the Z bit. This leaves the STATUS register as '000u u1uu' (where u = unchanged).

It is recommended, therefore, that only BCF, BSF, SWAPF and MOVWF instructions are used to alter the STATUS register, because these instructions do not affect any Status bits. For other instructions not affecting any Status bits, see **Section 15.0 "Instruction Set Summary"**

Note 1: The C and DC bits operate as a Borrow and Digit Borrow out bit, respectively, in subtraction.

REGISTER 2-1: STATUS: STATUS REGISTER

R/W-0	R/W-0	R/W-0	R-1	R-1	R/W-x	R/W-x	R/W-x
IRP	RP1	RP0	$\overline{\text{TO}}$	$\overline{\text{PD}}$	Z	DC ⁽¹⁾	C ⁽¹⁾
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

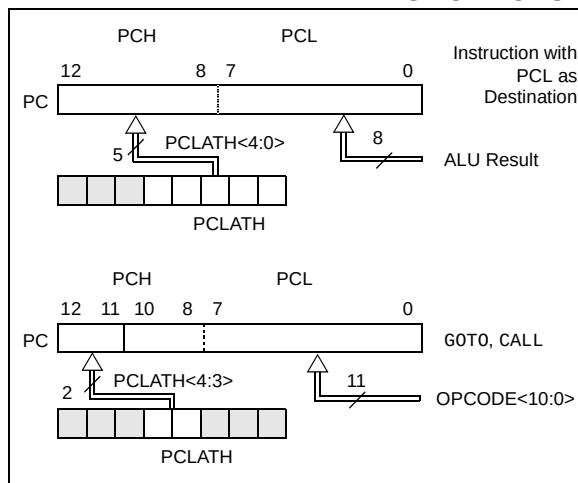
bit 7	IRP: Register Bank Select bit (used for indirect addressing) 1 = Bank 2, 3 (100h-1FFh) 0 = Bank 0, 1 (00h-FFh)
bit 6-5	RP<1:0>: Register Bank Select bits (used for direct addressing) 00 = Bank 0 (00h-7Fh) 01 = Bank 1 (80h-FFh) 10 = Bank 2 (100h-17Fh) 11 = Bank 3 (180h-1FFh)
bit 4	$\overline{\text{TO}}$: Time-out bit 1 = After power-up, CLRWDT instruction or SLEEP instruction 0 = A WDT time-out occurred
bit 3	$\overline{\text{PD}}$: Power-down bit 1 = After power-up or by the CLRWDT instruction 0 = By execution of the SLEEP instruction
bit 2	Z: Zero bit 1 = The result of an arithmetic or logic operation is zero 0 = The result of an arithmetic or logic operation is not zero
bit 1	DC: Digit Carry/Borrow bit (ADDWF, ADDLW, SUBLW, SUBWF instructions) ⁽¹⁾ 1 = A carry-out from the 4th low-order bit of the result occurred 0 = No carry-out from the 4th low-order bit of the result
bit 0	C: Carry/Borrow bit (ADDWF, ADDLW, SUBLW, SUBWF instructions) ⁽¹⁾ 1 = A carry-out from the Most Significant bit of the result occurred 0 = No carry-out from the Most Significant bit of the result occurred

Note 1: For Borrow, the polarity is reversed. A subtraction is executed by adding the two's complement of the second operand. For rotate (RRF, RLF) instructions, this bit is loaded with either the high-order or low-order bit of the source register.

2.3 PCL and PCLATH

The Program Counter (PC) is 13 bits wide. The low byte comes from the PCL register, which is a readable and writable register. The high byte (PC<12:8>) is not directly readable or writable and comes from PCLATH. On any Reset, the PC is cleared. Figure 2-7 shows the two situations for the loading of the PC. The upper example in Figure 2-7 shows how the PC is loaded on a write to PCL (PCLATH<4:0> → PCH). The lower example in Figure 2-7 shows how the PC is loaded during a CALL or GOTO instruction (PCLATH<4:3> → PCH).

FIGURE 2-7: LOADING OF PC IN DIFFERENT SITUATIONS



2.3.1 MODIFYING PCL

Executing any instruction with the PCL register as the destination simultaneously causes the Program Counter PC<12:8> bits (PCH) to be replaced by the contents of the PCLATH register. This allows the entire contents of the program counter to be changed by writing the desired upper 5 bits to the PCLATH register. When the lower 8 bits are written to the PCL register, all 13 bits of the program counter will change to the values contained in the PCLATH register and those being written to the PCL register.

A computed GOTO is accomplished by adding an offset to the program counter (ADDWF PCL). Care should be exercised when jumping into a look-up table or program branch table (computed GOTO) by modifying the PCL register. Assuming that PCLATH is set to the table start address, if the table length is greater than 255 instructions or if the lower 8 bits of the memory address rolls over from 0xFF to 0x00 in the middle of the table, then PCLATH must be incremented for each address rollover that occurs between the table beginning and the target location within the table.

For more information refer to Application Note AN556, "Implementing a Table Read" (DS00556).

2.3.2 STACK

The PIC16F882/883/884/886/887 devices have an 8-level x 13-bit wide hardware stack (see Figures 2-2 and 2-3). The stack space is not part of either program or data space and the Stack Pointer is not readable or writable. The PC is PUSHed onto the stack when a CALL instruction is executed or an interrupt causes a branch. The stack is POPed in the event of a RETURN, RETLW or a RETFIE instruction execution. PCLATH is not affected by a PUSH or POP operation.

The stack operates as a circular buffer. This means that after the stack has been PUSHed eight times, the ninth push overwrites the value that was stored from the first push. The tenth push overwrites the second push (and so on).

- Note 1:** There are no Status bits to indicate stack overflow or stack underflow conditions.
- 2:** There are no instructions/mnemonics called PUSH or POP. These are actions that occur from the execution of the CALL, RETURN, RETLW and RETFIE instructions or the vectoring to an interrupt address.

2.4 Indirect Addressing, INDF and FSR Registers

The INDF register is not a physical register. Addressing the INDF register will cause indirect addressing.

Indirect addressing is possible by using the INDF register. Any instruction using the INDF register actually accesses data pointed to by the File Select Register (FSR). Reading INDF itself indirectly will produce 00h. Writing to the INDF register indirectly results in a no operation (although Status bits may be affected). An effective 9-bit address is obtained by concatenating the 8-bit FSR and the IRP bit of the STATUS register, as shown in Figure 2-8.

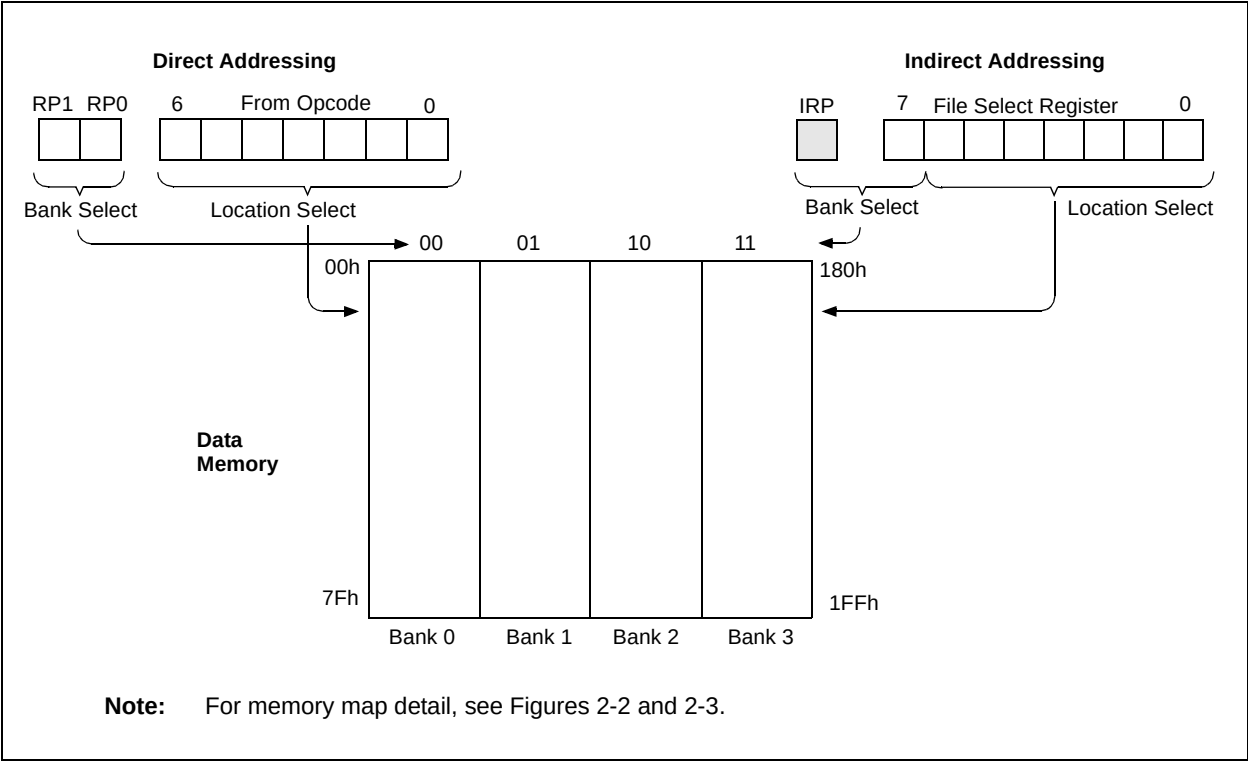
A simple program to clear RAM location 20h-2Fh using indirect addressing is shown in Example 2-1.

EXAMPLE 2-1: INDIRECT ADDRESSING

	MOVLW	0x20	;initialize pointer
	MOVWF	FSR	;to RAM
NEXT	CLRF	INDF	;clear INDF register
	INCF	FSR	;inc pointer
	BTFSS	FSR, 4	;all done?
	GOTO	NEXT	;no clear next
CONTINUE			;yes continue

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FIGURE 2-8: DIRECT/INDIRECT ADDRESSING PIC16F882/883/884/886/887



3.0 I/O PORTS

There are as many as thirty-five general purpose I/O pins available. Depending on which peripherals are enabled, some or all of the pins may not be available as general purpose I/O. In general, when a peripheral is enabled, the associated pin may not be used as a general purpose I/O pin.

3.1 PORTA and the TRISA Registers

PORTA is a 8-bit wide, bidirectional port. The corresponding data direction register is TRISA (Register 3-2). Setting a TRISA bit (= 1) will make the corresponding PORTA pin an input (i.e., disable the output driver). Clearing a TRISA bit (= 0) will make the corresponding PORTA pin an output (i.e., enables output driver and puts the contents of the output latch on the selected pin). Example 3-1 shows how to initialize PORTA.

Reading the PORTA register (Register 3-1) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write

operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

The TRISA register (Register 3-2) controls the PORTA pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISA register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

Note: The ANSEL register must be initialized to configure an analog channel as a digital input. Pins configured as analog inputs will read '0'.

EXAMPLE 3-1: INITIALIZING PORTA

```
BANKSEL PORTA      ;
CLRF   PORTA        ;Init PORTA
BANKSEL ANSEL       ;
CLRF   ANSEL         ;digital I/O
BANKSEL TRISA       ;
MOVLW  0Ch          ;Set RA<3:2> as inputs
MOVWF  TRISA         ;and set RA<5:4,1:0>
                      ;as outputs
```

REGISTER 3-1: PORTA: PORTA REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RA7	RA6	RA5	RA4	RA3	RA2	RA1	RA0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **RA<7:0>**: PORTA I/O Pin bit
 1 = Port pin is > V_{IH}
 0 = Port pin is < V_{IL}

REGISTER 3-2: TRISA: PORTA TRI-STATE REGISTER

R/W-1 ⁽¹⁾	R/W-1 ⁽¹⁾	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TRISA7	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **TRISA<7:0>**: PORTA Tri-State Control bit
 1 = PORTA pin configured as an input (tri-stated)
 0 = PORTA pin configured as an output

Note 1: TRISA<7:6> always reads '1' in XT, HS and LP Oscillator modes.

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3.2 Additional Pin Functions

RA0 also has an Ultra Low-Power Wake-up option. The next three sections describe these functions.

3.2.1 ANSEL REGISTER

The ANSEL register (Register 3-3) is used to configure the Input mode of an I/O pin to analog. Setting the appropriate ANSEL bit high will cause all digital reads on the pin to be read as '0' and allow analog functions on the pin to operate correctly.

The state of the ANSEL bits has no affect on digital output functions. A pin with TRIS clear and ANSEL set will still operate as a digital output, but the Input mode will be analog. This can cause unexpected behavior when executing read-modify-write instructions on the affected port.

REGISTER 3-3: ANSEL: ANALOG SELECT REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
ANS7 ⁽²⁾	ANS6 ⁽²⁾	ANS5 ⁽²⁾	ANS4	ANS3	ANS2	ANS1	ANS0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0

ANS<7:0>: Analog Select bits

Analog select between analog or digital function on pins AN<7:0>, respectively.

1 = Analog input. Pin is assigned as analog input⁽¹⁾.

0 = Digital I/O. Pin is assigned to port or special function.

Note 1: Setting a pin to an analog input automatically disables the digital input circuitry, weak pull-ups, and interrupt-on-change if available. The corresponding TRIS bit must be set to Input mode in order to allow external control of the voltage on the pin.

2: Not implemented on PIC16F883/886.

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3.2.3.8 RA7/OSC1/CLKIN

Figure 3-8 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- a crystal/resonator connection
- a clock input

FIGURE 3-8: BLOCK DIAGRAM OF RA7

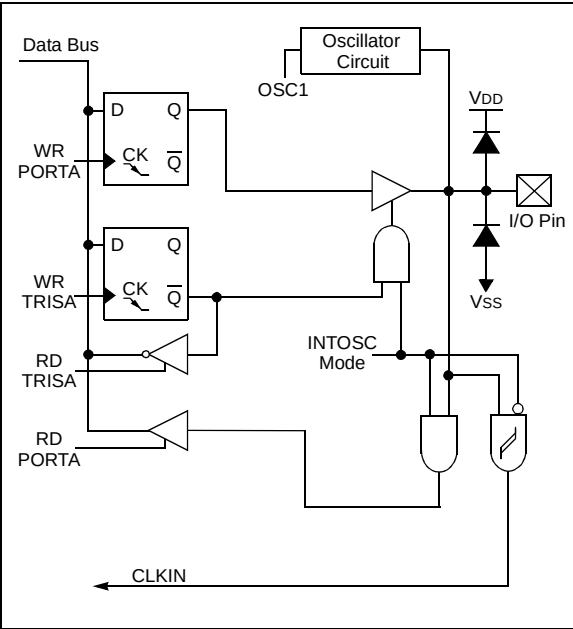


TABLE 3-1: SUMMARY OF REGISTERS ASSOCIATED WITH PORTA

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
ADCON0	ADCS1	ADCS0	CHS3	CHS2	CHS1	CHS0	GO/DONE	ADON	0000 0000	0000 0000
ANSEL	ANS7	ANS6	ANS5	ANS4	ANS3	ANS2	ANS1	ANS0	1111 1111	1111 1111
CM1CON0	C1ON	C1OUT	C1OE	C1POL	—	C1R	C1CH1	C1CH0	0000 -000	0000 -000
CM2CON0	C2ON	C2OUT	C2OE	C2POL	—	C2R	C2CH1	C2CH0	0000 -000	0000 -000
CM2CON1	MC1OUT	MC2OUT	C1RSEL	C2RSEL	—	—	T1GSS	C2SYNC	0000 --10	0000 --10
PCON	—	—	ULPWUE	SBOREN	—	—	POR	BOR	--01 --qq	--0u --uu
OPTION_REG	RBPU	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	1111 1111
PORTA	RA7	RA6	RA5	RA4	RA3	RA2	RA1	RA0	xxxx xxxx	uuuu uuuu
SSPCON	WCOL	SSPOV	SSPEN	CKP	SSPM3	SSPM2	SSPM1	SSPM0	0000 0000	0000 0000
TRISA	TRISA7	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, — = unimplemented locations read as '0'. Shaded cells are not used by PORTA.

3.3 PORTB and TRISB Registers

PORTB is an 8-bit wide, bidirectional port. The corresponding data direction register is TRISB (Register 3-6). Setting a TRISB bit (= 1) will make the corresponding PORTB pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISB bit (= 0) will make the corresponding PORTB pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). Example 3-3 shows how to initialize PORTB.

Reading the PORTB register (Register 3-5) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

The TRISB register (Register 3-6) controls the PORTB pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISB register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'. Example 3-3 shows how to initialize PORTB.

EXAMPLE 3-3: INITIALIZING PORTB

```
BANKSEL PORTB      ;
CLRF  PORTB        ;Init PORTB
BANKSEL TRISB       ;
MOVLW  B'11110000' ;Set RB<7:4> as inputs
                        ;and RB<3:0> as outputs
MOVWF  TRISB        ;
```

Note: The ANSELH register must be initialized to configure an analog channel as a digital input. Pins configured as analog inputs will read '0'.

3.4 Additional PORTB Pin Functions

PORTB pins RB<7:0> on the device family device have an interrupt-on-change option and a weak pull-up option. The following three sections describe these PORTB pin functions.

Every PORTB pin on this device family has an interrupt-on-change option and a weak pull-up option.

3.4.1 ANSELH REGISTER

The ANSELH register (Register 3-4) is used to configure the Input mode of an I/O pin to analog. Setting the appropriate ANSELH bit high will cause all digital reads on the pin to be read as '0' and allow analog functions on the pin to operate correctly.

The state of the ANSELH bits has no affect on digital output functions. A pin with TRIS clear and ANSELH set will still operate as a digital output, but the Input mode will be analog. This can cause unexpected behavior when executing read-modify-write instructions on the affected port.

3.4.2 WEAK PULL-UPS

Each of the PORTB pins has an individually configurable internal weak pull-up. Control bits WPUB<7:0> enable or disable each pull-up (see Register 3-7). Each weak pull-up is automatically turned off when the port pin is configured as an output. All pull-ups are disabled on a Power-on Reset by the RBPU bit of the OPTION register.

3.4.3 INTERRUPT-ON-CHANGE

All of the PORTB pins are individually configurable as an interrupt-on-change pin. Control bits IOCB<7:0> enable or disable the interrupt function for each pin. Refer to Register 3-8. The interrupt-on-change feature is disabled on a Power-on Reset.

For enabled interrupt-on-change pins, the present value is compared with the old value latched on the last read of PORTB to determine which bits have changed or mismatched the old value. The 'mismatch' outputs of the last read are OR'd together to set the PORTB Change Interrupt flag bit (RBIF) in the INTCON register.

This interrupt can wake the device from Sleep. The user, in the Interrupt Service Routine, clears the interrupt by:

- Any read or write of PORTB. This will end the mismatch condition.
- Clear the flag bit RBIF.

A mismatch condition will continue to set flag bit RBIF. Reading or writing PORTB will end the mismatch condition and allow flag bit RBIF to be cleared. The latch holding the last read value is not affected by a MCLR nor Brown-out Reset. After these Resets, the RBIF flag will continue to be set if a mismatch is present.

Note: If a change on the I/O pin should occur when the read operation is being executed (start of the Q2 cycle), then the RBIF interrupt flag may not get set. Furthermore, since a read or write on a port affects all bits of that port, care must be taken when using multiple pins in Interrupt-on-Change mode. Changes on one pin may not be seen while servicing changes on another pin.

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REGISTER 3-4: ANSELH: ANALOG SELECT HIGH REGISTER

U-0	U-0	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
—	—	ANS13	ANS12	ANS11	ANS10	ANS9	ANS8
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-6 **Unimplemented:** Read as '0'

bit 5-0 **ANS<13:8>:** Analog Select bits

Analog select between analog or digital function on pins AN<13:8>, respectively.

1 = Analog input. Pin is assigned as analog input⁽¹⁾.

0 = Digital I/O. Pin is assigned to port or special function.

Note 1: Setting a pin to an analog input automatically disables the digital input circuitry, weak pull-ups, and interrupt-on-change if available. The corresponding TRIS bit must be set to Input mode in order to allow external control of the voltage on the pin.

REGISTER 3-5: PORTB: PORTB REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **RB<7:0>:** PORTB I/O Pin bit

1 = Port pin is > V_{IH}

0 = Port pin is < V_{IL}

REGISTER 3-6: TRISB: PORTB TRI-STATE REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **TRISB<7:0>:** PORTB Tri-State Control bit

1 = PORTB pin configured as an input (tri-stated)

0 = PORTB pin configured as an output

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3.4.4 PIN DESCRIPTIONS AND DIAGRAMS

Each PORTB pin is multiplexed with other functions. The pins and their combined functions are briefly described here. For specific information about individual functions such as the SSP, I²C or interrupts, refer to the appropriate section in this data sheet.

3.4.4.1 RB0/AN12/INT

Figure 3-9 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC
- an external edge triggered interrupt

3.4.4.2 RB1/AN10/P1C⁽¹⁾/C12IN3-

Figure 3-9 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC
- a PWM output⁽¹⁾
- an analog input to Comparator C1 or C2

Note 1: P1C is available on PIC16F882/883/886 only.

3.4.4.3 RB2/AN8/P1B⁽¹⁾

Figure 3-9 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC
- a PWM output⁽¹⁾

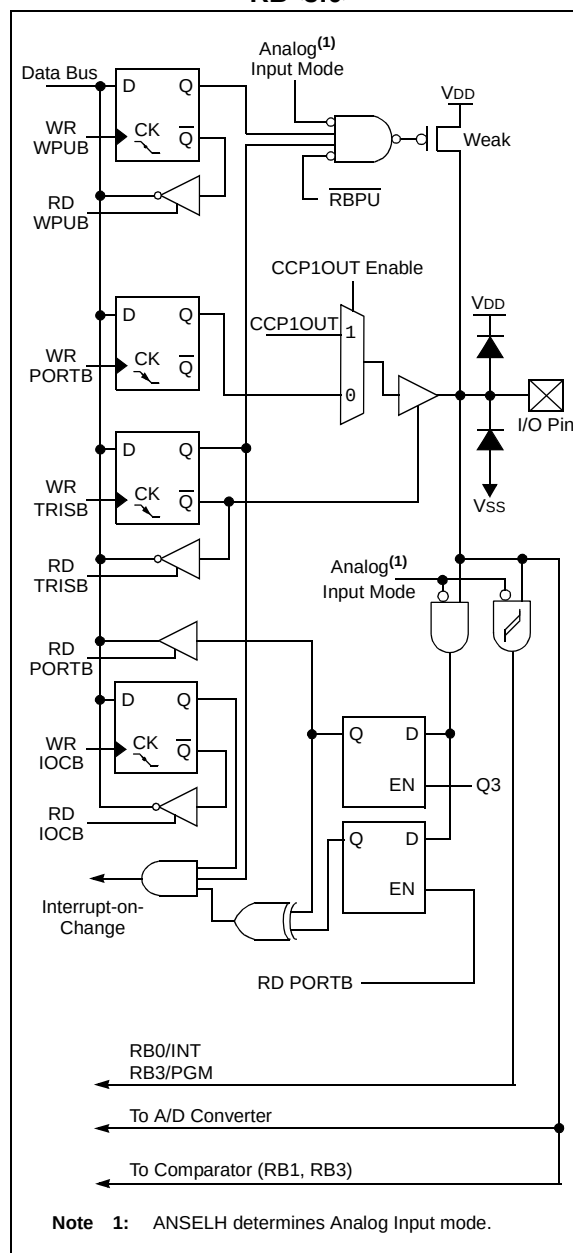
Note 1: P1B is available on PIC16F882/883/886 only.

3.4.4.4 RB3/AN9/PGM/C12IN2-

Figure 3-9 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC
- Low-voltage In-Circuit Serial Programming enable pin
- an analog input to Comparator C1 or C2

FIGURE 3-9: BLOCK DIAGRAM OF RB<3:0>



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TABLE 3-2: SUMMARY OF REGISTERS ASSOCIATED WITH PORTB

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
ANSELH	—	—	ANS13	ANS12	ANS11	ANS10	ANS9	ANS8	--11 1111	--11 1111
CCP1CON	P1M1	P1M0	DC1B1	DC1B0	CCP1M3	CCP1M2	CCP1M1	CCP1M0	0000 0000	0000 0000
CM2CON1	MC1OUT	MC2OUT	C1RSEL	C2RSEL	—	—	T1GSS	C2SYNC	0000 --10	0000 --10
IOCB	IOCB7	IOCB6	IOCB5	IOCB4	IOCB3	IOCB2	IOCB1	IOCB0	0000 0000	0000 0000
INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	0000 000x	0000 000x
OPTION_REG	RBP $\overline{\text{U}}$	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	1111 1111	1111 1111
PORTB	RB7	RB6	RB5	RB4	RB3	RB2	RB1	RB0	xxxx xxxx	uuuu uuuu
TRISB	TRISB7	TRISB6	TRISB5	TRISB4	TRISB3	TRISB2	TRISB1	TRISB0	1111 1111	1111 1111
WPUB	WPUB7	WPUB6	WPUB5	WPUB4	WPUB3	WPUB2	WPUB1	WPUB0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, — = unimplemented read as '0'. Shaded cells are not used by PORTB.

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3.5 PORTC and TRISC Registers

PORTC is a 8-bit wide, bidirectional port. The corresponding data direction register is TRISC (Register 3-10). Setting a TRISC bit (= 1) will make the corresponding PORTC pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISC bit (= 0) will make the corresponding PORTC pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). Example 3-4 shows how to initialize PORTC.

Reading the PORTC register (Register 3-9) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

The TRISC register (Register 3-10) controls the PORTC pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISC register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

EXAMPLE 3-4: INITIALIZING PORTC

```
BANKSEL PORTC      ;
CLRWF  PORTC        ;Init PORTC
BANKSEL TRISC       ;
MOVLW  B'00001100' ;Set RC<3:2> as inputs
MOVWF  TRISC         ;and set RC<7:4,1:0>
                        ;as outputs
```

REGISTER 3-9: PORTC: PORTC REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RC7	RC6	RC5	RC4	RC3	RC2	RC1	RC0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **RC<7:0>**: PORTC General Purpose I/O Pin bit

1 = Port pin is > V_{IH}

0 = Port pin is < V_{IL}

REGISTER 3-10: TRISC: PORTC TRI-STATE REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1 ⁽¹⁾	R/W-1 ⁽¹⁾
TRISC7	TRISC6	TRISC5	TRISC4	TRISC3	TRISC2	TRISC1	TRISC0
bit 7							bit 0

Legend:

R = Readable bit

W = Writable bit

U = Unimplemented bit, read as '0'

-n = Value at POR

'1' = Bit is set

'0' = Bit is cleared

x = Bit is unknown

bit 7-0 **TRISC<7:0>**: PORTC Tri-State Control bit

1 = PORTC pin configured as an input (tri-stated)

0 = PORTC pin configured as an output

Note 1: TRISC<1:0> always reads '1' in LP Oscillator mode.

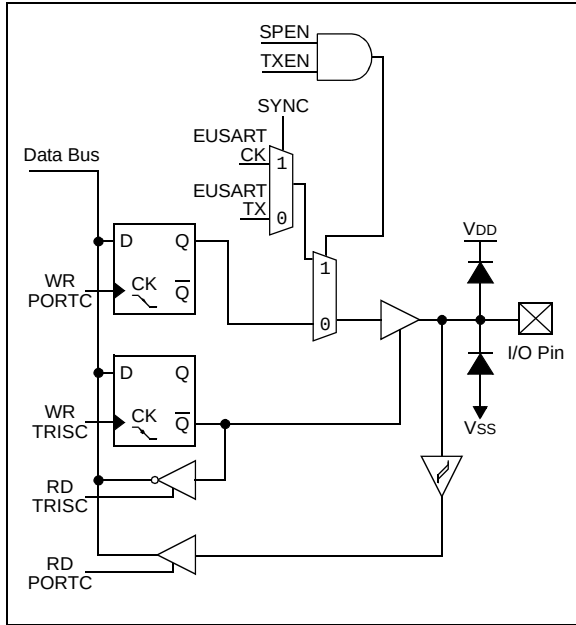
PIC16F882/883/884/886/887

3.5.7 RC6/TX/CK

Figure 3-17 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- an asynchronous serial output
- a synchronous clock I/O

FIGURE 3-17: BLOCK DIAGRAM OF RC6



3.5.8 RC7/RX/DT

Figure 3-18 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose I/O
- an asynchronous serial input
- a synchronous serial data I/O

FIGURE 3-18: BLOCK DIAGRAM OF RC7

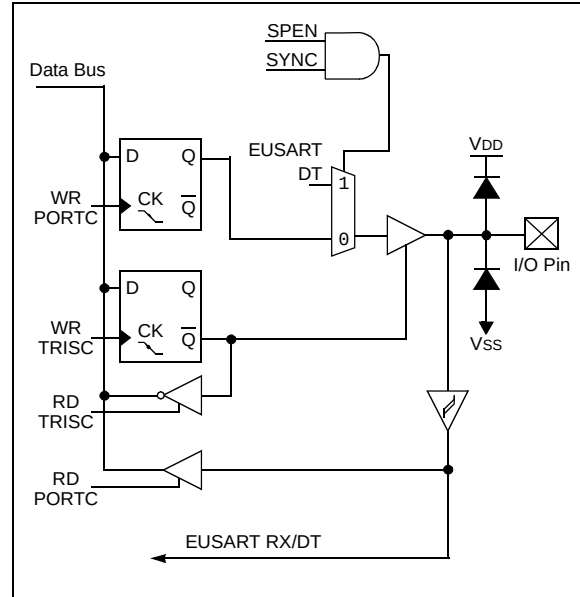


TABLE 3-3: SUMMARY OF REGISTERS ASSOCIATED WITH PORTC

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
CCP1CON	P1M1	P1M0	DC1B1	DC1B0	CCP1M3	CCP1M2	CCP1M1	CCP1M0	0000 0000	0000 0000
CCP2CON	—	—	DC2B1	DC2B0	CCP2M3	CCP2M2	CCP2M1	CCP2M0	--00 0000	--00 0000
PORTC	RC7	RC6	RC5	RC4	RC3	RC2	RC1	RC0	xxxx xxxx	uuuu uuuu
PSTRCON	—	—	—	STRSYNC	STRD	STRC	STRB	STRA	--0 0001	--0 0001
RCSTA	SPEN	RX9	SREN	CREN	ADDEN	FERR	OERR	RX9D	0000 000x	0000 000x
SSPCON	WCOL	SSPOV	SSPEN	CKP	SSPM3	SSPM2	SSPM1	SSPM0	0000 0000	0000 0000
T1CON	T1GINV	TMR1GE	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON	0000 0000	0000 0000
TRISC	TRISC7	TRISC6	TRISC5	TRISC4	TRISC3	TRISC2	TRISC1	TRISC0	1111 1111	1111 1111

Legend: x = unknown, u = unchanged, – = unimplemented locations read as '0'. Shaded cells are not used by PORTC.

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3.6 PORTD and TRISD Registers

PORTD⁽¹⁾ is a 8-bit wide, bidirectional port. The corresponding data direction register is TRISD (Register 3-12). Setting a TRISD bit (= 1) will make the corresponding PORTD pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISD bit (= 0) will make the corresponding PORTD pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). Example 3-5 shows how to initialize PORTD.

Reading the PORTD register (Register 3-11) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch.

Note 1: PORTD is available on PIC16F884/887 only.

The TRISD register (Register 3-12) controls the PORTD pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISD register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

EXAMPLE 3-5: INITIALIZING PORTD

```
BANKSEL PORTD      ;
CLRF   PORTD        ;Init PORTD
BANKSEL TRISD       ;
MOVLW  B'00001100' ;Set RD<3:2> as inputs
MOVWF  TRISD        ;and set RD<7:4,1:0>
                        ;as outputs
```

REGISTER 3-11: PORTD: PORTD REGISTER

R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x	R/W-x
RD7	RD6	RD5	RD4	RD3	RD2	RD1	RD0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
-n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **RD<7:0>**: PORTD General Purpose I/O Pin bit
1 = Port pin is > V_{IH}
0 = Port pin is < V_{IL}

REGISTER 3-12: TRISD: PORTD TRI-STATE REGISTER

R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1	R/W-1
TRISD7	TRISD6	TRISD5	TRISD4	TRISD3	TRISD2	TRISD1	TRISD0
bit 7							bit 0

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
-n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-0 **TRISD<7:0>**: PORTD Tri-State Control bit
1 = PORTD pin configured as an input (tri-stated)
0 = PORTD pin configured as an output

3.7 PORTE and TRISE Registers

PORTE⁽¹⁾ is a 4-bit wide, bidirectional port. The corresponding data direction register is TRISE. Setting a TRISE bit (= 1) will make the corresponding PORTE pin an input (i.e., put the corresponding output driver in a High-Impedance mode). Clearing a TRISE bit (= 0) will make the corresponding PORTE pin an output (i.e., enable the output driver and put the contents of the output latch on the selected pin). The exception is RE3, which is input only and its TRIS bit will always read as '1'. Example 3-6 shows how to initialize PORTE.

Reading the PORTE register (Register 3-13) reads the status of the pins, whereas writing to it will write to the PORT latch. All write operations are read-modify-write operations. Therefore, a write to a port implies that the port pins are read, this value is modified and then written to the PORT data latch. RE3 reads '0' when MCLR = 1.

Note 1: RE<2:0> pins are available on PIC16F884/887 only.

The TRISE register (Register 3-14) controls the PORTE pin output drivers, even when they are being used as analog inputs. The user should ensure the bits in the TRISE register are maintained set when using them as analog inputs. I/O pins configured as analog input always read '0'.

Note: The ANSEL register must be initialized to configure an analog channel as a digital input. Pins configured as analog inputs will read '0'.

EXAMPLE 3-6: INITIALIZING PORTE

```
BANKSEL PORTE      ;
CLRF   PORTE       ;Init PORTE
BANKSEL ANSEL      ;
CLRF   ANSEL       ;digital I/O
BCF    STATUS,RP1  ;Bank 1
BANKSEL TRISE      ;
MOVLW  B'00001100' ;Set RE<3:2> as inputs
MOVWF  TRISE       ;and set RE<1:0>
                        ;as outputs
```

REGISTER 3-13: PORTE: PORTE REGISTER

U-0	U-0	U-0	U-0	R-x	R/W-x	R/W-x	R/W-x
—	—	—	—	RE3	RE2	RE1	RE0
bit 7				bit 0			

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-4 **Unimplemented:** Read as '0'
 bit 3-0 **RD<3:0>:** PORTE General Purpose I/O Pin bit
 1 = Port pin is > V_{IH}
 0 = Port pin is < V_{IL}

REGISTER 3-14: TRISE: PORTE TRI-STATE REGISTER

U-0	U-0	U-0	U-0	R-1 ⁽¹⁾	R/W-1	R/W-1	R/W-1
—	—	—	—	TRISE3	TRISE2	TRISE1	TRISE0
bit 7				bit 0			

Legend:

R = Readable bit W = Writable bit U = Unimplemented bit, read as '0'
 -n = Value at POR '1' = Bit is set '0' = Bit is cleared x = Bit is unknown

bit 7-4 **Unimplemented:** Read as '0'
 bit 3-0 **TRISE<3:0>:** PORTE Tri-State Control bit
 1 = PORTE pin configured as an input (tri-stated)
 0 = PORTE pin configured as an output

Note 1: TRISE<3> always reads '1'.

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3.7.1 RE0/AN5⁽¹⁾

This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC

Note 1: RE0/AN5 is available on PIC16F884/887 only.

3.7.2 RE1/AN6⁽¹⁾

This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC

Note 1: RE1/AN6 is available on PIC16F884/887 only.

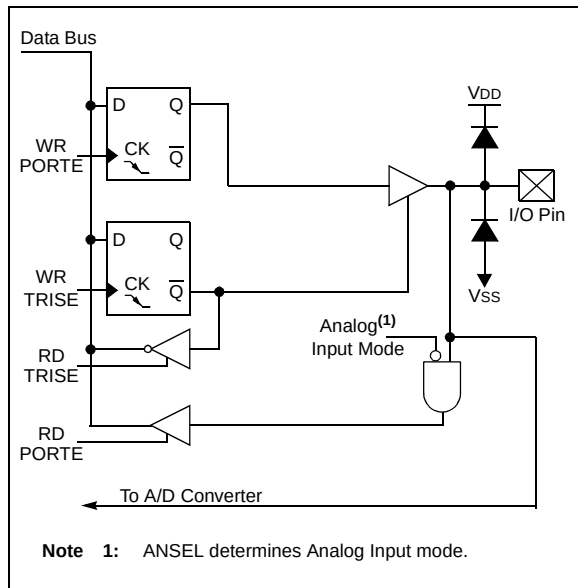
3.7.3 RE2/AN7⁽¹⁾

This pin is configurable to function as one of the following:

- a general purpose I/O
- an analog input for the ADC

Note 1: RE2/AN7 is available on PIC16F884/887 only.

FIGURE 3-21: BLOCK DIAGRAM OF RE<2:0>



3.7.4 RE3/MCLR/VPP

Figure 3-22 shows the diagram for this pin. This pin is configurable to function as one of the following:

- a general purpose input
- as Master Clear Reset with weak pull-up

FIGURE 3-22: BLOCK DIAGRAM OF RE3

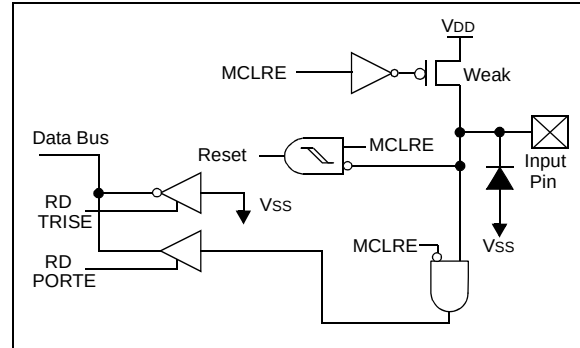


TABLE 3-5: SUMMARY OF REGISTERS ASSOCIATED WITH PORTE

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
ANSEL	ANS7	ANS6	ANS5	ANS4	ANS3	ANS2	ANS1	ANS0	1111 1111	1111 1111
PORTE	—	—	—	—	RE3	RE2	RE1	RE0	---- xxxx	---- uuuu
TRISE	—	—	—	—	TRISE3	TRISE2	TRISE1	TRISE0	---- 1111	---- 1111

Legend: x = unknown, u = unchanged, – = unimplemented locations read as '0'. Shaded cells are not used by PORTE

TABLE 14-4: INITIALIZATION CONDITION FOR REGISTER

Register	Address	Power-on Reset	MCLR Reset WDT Reset Brown-out Reset ⁽¹⁾	Wake-up from Sleep through Interrupt Wake-up from Sleep through WDT Time-out
W	—	xxxx xxxx	uuuu uuuu	uuuu uuuu
INDF	00h/80h/100h/180h	xxxx xxxx	xxxx xxxx	uuuu uuuu
TMR0	01h/101h	xxxx xxxx	uuuu uuuu	uuuu uuuu
PCL	02h/82h/102h/182h	0000 0000	0000 0000	PC + 1 ⁽³⁾
STATUS	03h/83h/103h/183h	0001 1xxx	000q quuu ⁽⁴⁾	uuuq quuu ⁽⁴⁾
FSR	04h/84h/104h/184h	xxxx xxxx	uuuu uuuu	uuuu uuuu
PORTA	05h	xxxx xxxx	0000 0000	uuuu uuuu
PORTB	06h/106h	xxxx xxxx	0000 0000	uuuu uuuu
PORTC	07h	xxxx xxxx	0000 0000	uuuu uuuu
PORTD	08h	xxxx xxxx	0000 0000	uuuu uuuu
PORTE	09h	---- xxxx	---- 0000	---- uuuu
PCLATH	0Ah/8Ah/10Ah/18Ah	---0 0000	---0 0000	---u uuuu
INTCON	0Bh/8Bh/10Bh/18Bh	0000 000x	0000 000u	uuuu uuuu ⁽²⁾
PIR1	0Ch	0000 0000	0000 0000	uuuu uuuu ⁽²⁾
PIR2	0Dh	0000 0000	0000 0000	uuuu uuuu ⁽²⁾
TMR1L	0Eh	xxxx xxxx	uuuu uuuu	uuuu uuuu
TMR1H	0Fh	xxxx xxxx	uuuu uuuu	uuuu uuuu
T1CON	10h	0000 0000	uuuu uuuu	-uuu uuuu
TMR2	11h	0000 0000	0000 0000	uuuu uuuu
T2CON	12h	-000 0000	-000 0000	-uuu uuuu
SSPBUF	13h	xxxx xxxx	uuuu uuuu	uuuu uuuu
SSPCON	14h	0000 0000	0000 0000	uuuu uuuu
CCPR1L	15h	xxxx xxxx	uuuu uuuu	uuuu uuuu
CCPR1H	16h	xxxx xxxx	uuuu uuuu	uuuu uuuu
CCP1CON	17h	0000 0000	0000 0000	uuuu uuuu
RCSTA	18h	0000 000x	0000 0000	uuuu uuuu
TXREG	19h	0000 0000	0000 0000	uuuu uuuu
RCREG	1Ah	0000 0000	0000 0000	uuuu uuuu
CCPR2L	1Bh	xxxx xxxx	uuuu uuuu	uuuu uuuu

Legend: u = unchanged, x = unknown, – = unimplemented bit, reads as '0', q = value depends on condition.

Note 1: If VDD goes too low, Power-on Reset will be activated and registers will be affected differently.

2: One or more bits in INTCON and/or PIR1 will be affected (to cause wake-up).

3: When the wake-up is due to an interrupt and the GIE bit is set, the PC is loaded with the interrupt vector (0004h).

4: See Table 14-5 for Reset value for specific condition.

5: If Reset was due to brown-out, then bit 0 = 0. All other Resets will cause bit 0 = u.

6: Accessible only when SSPCON register bits SSPM<3:0> = 1001.

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TABLE 14-4: INITIALIZATION CONDITION FOR REGISTER (CONTINUED)

Register	Address	Power-on Reset	MCLR Reset WDT Reset (Continued) Brown-out Reset ⁽¹⁾	Wake-up from Sleep through Interrupt Wake-up from Sleep through WDT Time-out (Continued)
CCPR2H	1Ch	xxxx xxxx	uuuu uuuu	uuuu uuuu
CCP2CON	1Dh	--00 0000	--00 0000	--uu uuuu
ADRESH	1Eh	xxxx xxxx	uuuu uuuu	uuuu uuuu
ADCON0	1Fh	00-0 0000	00-0 0000	uu-u uuuu
OPTION_REG	81h/181h	1111 1111	1111 1111	uuuu uuuu
TRISA	85h	1111 1111	1111 1111	uuuu uuuu
TRISB	86h/186h	1111 1111	1111 1111	uuuu uuuu
TRISC	87h	1111 1111	1111 1111	uuuu uuuu
TRISD	88h	1111 1111	1111 1111	uuuu uuuu
TRISE	89h	---- 1111	---- 1111	---- uuuu
PIE1	8Ch	0000 0000	0000 0000	uuuu uuuu
PIE2	8Dh	0000 0000	0000 0000	uuuu uuuu
PCON	8Eh	--01 --0x	--0u --uu ^(1, 5)	--uu --uu
OSCCON	8Fh	-110 q000	-110 q000	-uuu uuuu
OSCTUNE	90h	---0 0000	---u uuuu	---u uuuu
SSPCON2	91h	0000 0000	0000 0000	uuuu uuuu
PR2	92h	1111 1111	1111 1111	1111 1111
SSPADD ⁽⁶⁾	93h	0000 0000	0000 0000	uuuu uuuu
SSPMSK ⁽⁶⁾	93h	1111 1111	1111 1111	1111 1111
SSPSTAT	94h	0000 0000	0000 0000	uuuu uuuu
WPUB	95h	1111 1111	1111 1111	uuuu uuuu
IOCB	96h	0000 0000	0000 0000	uuuu uuuu
VRCON	97h	0000 0000	0000 0000	uuuu uuuu
TXSTA	98h	0000 -010	0000 -010	uuuu -uuu
SPBRG	99h	0000 0000	0000 0000	uuuu uuuu
SPBRGH	9Ah	0000 0000	0000 0000	uuuu uuuu
PWM1CON	9Bh	0000 0000	0000 0000	uuuu uuuu
ECCPAS	9Ch	0000 0000	0000 0000	uuuu uuuu
PSTRCON	9Dh	---0 0001	---0 0001	---u uuuu
ADRESL	9Eh	xxxx xxxx	uuuu uuuu	uuuu uuuu
ADCON1	9Fh	0-00 ----	0-00 ----	u-uu ----
WDTCON	105h	---0 1000	---0 1000	---u uuuu
CM1CON0	107h	0000 0-00	0000 0-00	uuuu u-uu
CM2CON0	108h	0000 0-00	0000 0-00	uuuu u-uu

Legend: u = unchanged, x = unknown, – = unimplemented bit, reads as '0', q = value depends on condition.

Note 1: If VDD goes too low, Power-on Reset will be activated and registers will be affected differently.

2: One or more bits in INTCON and/or PIR1 will be affected (to cause wake-up).

3: When the wake-up is due to an interrupt and the GIE bit is set, the PC is loaded with the interrupt vector (0004h).

4: See Table 14-5 for Reset value for specific condition.

5: If Reset was due to brown-out, then bit 0 = 0. All other Resets will cause bit 0 = u.

6: Accessible only when SSPCON register bits SSPM<3:0> = 1001.

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TABLE 14-4: INITIALIZATION CONDITION FOR REGISTER (CONTINUED)

Register	Address	Power-on Reset	MCLR Reset WDT Reset (Continued) Brown-out Reset ⁽¹⁾	Wake-up from Sleep through Interrupt Wake-up from Sleep through WDT Time-out (Continued)
CM2CON1	109h	0000 0--0	0000 0--0	uuuu u--u
EEDAT	10Ch	0000 0000	0000 0000	uuuu uuuu
EEADR	10Dh	0000 0000	0000 0000	uuuu uuuu
EEDATH	10Eh	--00 0000	--00 0000	--uu uuuu
EEADRH	10Fh	---0 0000	---0 0000	---u uuuu
SRCON	185h	0000 00-0	0000 00-0	uuuu uu-u
BAUDCTL	187h	01-0 0-00	01-0 0-00	uu-u u-uu
ANSEL	188h	1111 1111	1111 1111	uuuu uuuu
ANSELH	189h	1111 1111	1111 1111	uuuu uuuu
EECON1	18Ch	---- x000	---- q000	---- uuuu
EECON2	18Dh	---- ----	---- ----	---- ----

Legend: u = unchanged, x = unknown, – = unimplemented bit, reads as '0', q = value depends on condition.

Note 1: If VDD goes too low, Power-on Reset will be activated and registers will be affected differently.

2: One or more bits in INTCON and/or PIR1 will be affected (to cause wake-up).

3: When the wake-up is due to an interrupt and the GIE bit is set, the PC is loaded with the interrupt vector (0004h).

4: See Table 14-5 for Reset value for specific condition.

5: If Reset was due to brown-out, then bit 0 = 0. All other Resets will cause bit 0 = u.

6: Accessible only when SSPCON register bits SSPM<3:0> = 1001.

TABLE 14-5: INITIALIZATION CONDITION FOR SPECIAL REGISTERS

Condition	Program Counter	Status Register	PCON Register
Power-on Reset	000h	0001 1xxx	--01 --0x
MCLR Reset during normal operation	000h	000u uuuu	--0u --uu
MCLR Reset during Sleep	000h	0001 0uuu	--0u --uu
WDT Reset	000h	0000 uuuu	--0u --uu
WDT Wake-up	PC + 1	uuu0 0uuu	--uu --uu
Brown-out Reset	000h	0001 1uuu	--01 --u0
Interrupt Wake-up from Sleep	PC + 1 ⁽¹⁾	uuu1 0uuu	--uu --uu

Legend: u = unchanged, x = unknown, – = unimplemented bit, reads as '0'.

Note 1: When the wake-up is due to an interrupt and Global Interrupt Enable bit, GIE, is set, the PC is loaded with the interrupt vector (0004h) after execution of PC + 1.

15.0 INSTRUCTION SET SUMMARY

The PIC16F882/883/884/886/887 instruction set is highly orthogonal and is comprised of three basic categories:

- **Byte-oriented** operations
- **Bit-oriented** operations
- **Literal and control** operations

Each PIC16 instruction is a 14-bit word divided into an **opcode**, which specifies the instruction type and one or more **operands**, which further specify the operation of the instruction. The formats for each of the categories is presented in Figure 15-1, while the various opcode fields are summarized in Table 15-1.

Table 15-2 lists the instructions recognized by the MPASM™ assembler.

For **byte-oriented** instructions, 'f' represents a file register designator and 'd' represents a destination designator. The file register designator specifies which file register is to be used by the instruction.

The destination designator specifies where the result of the operation is to be placed. If 'd' is zero, the result is placed in the W register. If 'd' is one, the result is placed in the file register specified in the instruction.

For **bit-oriented** instructions, 'b' represents a bit field designator, which selects the bit affected by the operation, while 'f' represents the address of the file in which the bit is located.

For **literal and control** operations, 'k' represents an 8-bit or 11-bit constant, or literal value.

One instruction cycle consists of four oscillator periods; for an oscillator frequency of 4 MHz, this gives a normal instruction execution time of 1 μs. All instructions are executed within a single instruction cycle, unless a conditional test is true, or the program counter is changed as a result of an instruction. When this occurs, the execution takes two instruction cycles, with the second cycle executed as a NOP.

All instruction examples use the format '0xhh' to represent a hexadecimal number, where 'h' signifies a hexadecimal digit.

15.1 Read-Modify-Write Operations

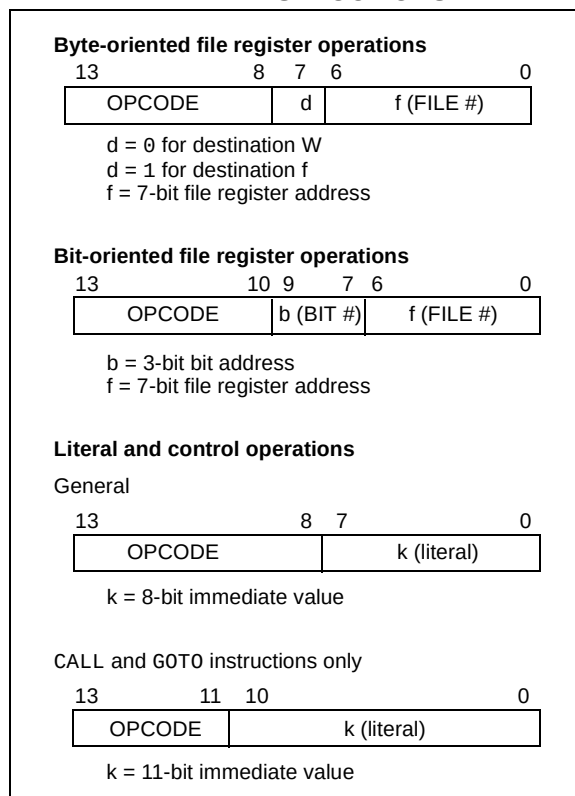
Any instruction that specifies a file register as part of the instruction performs a Read-Modify-Write (RMW) operation. The register is read, the data is modified, and the result is stored according to either the instruction, or the destination designator 'd'. A read operation is performed on a register even if the instruction writes to that register.

For example, a CLRF PORTA instruction will read PORTA, clear all the data bits, then write the result back to PORTA. This example would have the unintended consequence of clearing the condition that set the RAIF flag.

TABLE 15-1: OPCODE FIELD DESCRIPTIONS

Field	Description
f	Register file address (0x00 to 0x7F)
W	Working register (accumulator)
b	Bit address within an 8-bit file register
k	Literal field, constant data or label
x	Don't care location (= 0 or 1). The assembler will generate code with x = 0. It is the recommended form of use for compatibility with all Microchip software tools.
d	Destination select; d = 0: store result in W, d = 1: store result in file register f. Default is d = 1.
PC	Program Counter
TO	Time-out bit
C	Carry bit
DC	Digit carry bit
Z	Zero bit
PD	Power-down bit

FIGURE 15-1: GENERAL FORMAT FOR INSTRUCTIONS



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TABLE 15-2: PIC16F882/883/884/886/887 INSTRUCTION SET

Mnemonic, Operands		Description	Cycles	14-Bit Opcode				Status Affected	Notes
				MSb		LSb			
BYTE-ORIENTED FILE REGISTER OPERATIONS									
ADDWF	f, d	Add W and f	1	00	0111	dfff	ffff	C, DC, Z	1, 2
ANDWF	f, d	AND W with f	1	00	0101	dfff	ffff	Z	1, 2
CLRF	f	Clear f	1	00	0001	lfff	ffff	Z	2
CLRWF	—	Clear W	1	00	0001	0xxx	xxxx	Z	
COMF	f, d	Complement f	1	00	1001	dfff	ffff	Z	1, 2
DECF	f, d	Decrement f	1	00	0011	dfff	ffff	Z	1, 2
DECFSZ	f, d	Decrement f, Skip if 0	1(2)	00	1011	dfff	ffff		1, 2, 3
INCF	f, d	Increment f	1	00	1010	dfff	ffff	Z	1, 2
INCFSZ	f, d	Increment f, Skip if 0	1(2)	00	1111	dfff	ffff		1, 2, 3
IORWF	f, d	Inclusive OR W with f	1	00	0100	dfff	ffff	Z	1, 2
MOVF	f, d	Move f	1	00	1000	dfff	ffff	Z	1, 2
MOVWF	f	Move W to f	1	00	0000	lfff	ffff		
NOP	—	No Operation	1	00	0000	0xx0	0000		
RLF	f, d	Rotate Left f through Carry	1	00	1101	dfff	ffff	C	1, 2
RRF	f, d	Rotate Right f through Carry	1	00	1100	dfff	ffff	C	1, 2
SUBWF	f, d	Subtract W from f	1	00	0010	dfff	ffff	C, DC, Z	1, 2
SWAPF	f, d	Swap nibbles in f	1	00	1110	dfff	ffff		1, 2
XORWF	f, d	Exclusive OR W with f	1	00	0110	dfff	ffff	Z	1, 2
BIT-ORIENTED FILE REGISTER OPERATIONS									
BCF	f, b	Bit Clear f	1	01	00bb	bfff	ffff		1, 2
BSF	f, b	Bit Set f	1	01	01bb	bfff	ffff		1, 2
BTFSC	f, b	Bit Test f, Skip if Clear	1 (2)	01	10bb	bfff	ffff		3
BTFSS	f, b	Bit Test f, Skip if Set	1 (2)	01	11bb	bfff	ffff		3
LITERAL AND CONTROL OPERATIONS									
ADDLW	k	Add literal and W	1	11	111x	kkkk	kkkk	C, DC, Z	
ANDLW	k	AND literal with W	1	11	1001	kkkk	kkkk	Z	
CALL	k	Call Subroutine	2	10	0kkk	kkkk	kkkk		
CLRWDI	—	Clear Watchdog Timer	1	00	0000	0110	0100	$\overline{TO}$, $\overline{PD}$	
GOTO	k	Go to address	2	10	1kkk	kkkk	kkkk		
IORLW	k	Inclusive OR literal with W	1	11	1000	kkkk	kkkk	Z	
MOVLW	k	Move literal to W	1	11	00xx	kkkk	kkkk		
RETFIE	—	Return from interrupt	2	00	0000	0000	1001		
RETLW	k	Return with literal in W	2	11	01xx	kkkk	kkkk		
RETURN	—	Return from Subroutine	2	00	0000	0000	1000		
SLEEP	—	Go into Standby mode	1	00	0000	0110	0011	$\overline{TO}$, $\overline{PD}$	
SUBLW	k	Subtract W from literal	1	11	110x	kkkk	kkkk	C, DC, Z	
XORLW	k	Exclusive OR literal with W	1	11	1010	kkkk	kkkk	Z	

- Note 1:** When an I/O register is modified as a function of itself (e.g., MOVF GPIO, 1), the value used will be that value present on the pins themselves. For example, if the data latch is '1' for a pin configured as input and is driven low by an external device, the data will be written back with a '0'.
- 2:** If this instruction is executed on the TMR0 register (and where applicable, d = 1), the prescaler will be cleared if assigned to the Timer0 module.
- 3:** If the Program Counter (PC) is modified, or a conditional test is true, the instruction requires two cycles. The second cycle is executed as a NOP.

15.2 Instruction Descriptions

ADDLW Add literal and W

Syntax: [*label*] ADDLW *k*

Operands: $0 \leq k \leq 255$

Operation: $(W) + k \rightarrow (W)$

Status Affected: C, DC, Z

Description: The contents of the W register are added to the eight-bit literal 'k' and the result is placed in the W register.

BCF Bit Clear f

Syntax: [*label*] BCF *f*,*b*

Operands: $0 \leq f \leq 127$
 $0 \leq b \leq 7$

Operation: $0 \rightarrow (f)$

Status Affected: None

Description: Bit 'b' in register 'f' is cleared.

ADDWF Add W and f

Syntax: [*label*] ADDWF *f*,*d*

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(W) + (f) \rightarrow (\text{destination})$

Status Affected: C, DC, Z

Description: Add the contents of the W register with register 'f'. If 'd' is '0', the result is stored in the W register. If 'd' is '1', the result is stored back in register 'f'.

BSF Bit Set f

Syntax: [*label*] BSF *f*,*b*

Operands: $0 \leq f \leq 127$
 $0 \leq b \leq 7$

Operation: $1 \rightarrow (f)$

Status Affected: None

Description: Bit 'b' in register 'f' is set.

ANDLW AND literal with W

Syntax: [*label*] ANDLW *k*

Operands: $0 \leq k \leq 255$

Operation: $(W) .AND. (k) \rightarrow (W)$

Status Affected: Z

Description: The contents of W register are AND'ed with the eight-bit literal 'k'. The result is placed in the W register.

BTFSC Bit Test f, Skip if Clear

Syntax: [*label*] BTFSC *f*,*b*

Operands: $0 \leq f \leq 127$
 $0 \leq b \leq 7$

Operation: skip if $(f) = 0$

Status Affected: None

Description: If bit 'b' in register 'f' is '1', the next instruction is executed.
If bit 'b' in register 'f' is '0', the next instruction is discarded, and a NOP is executed instead, making this a two-cycle instruction.

ANDWF AND W with f

Syntax: [*label*] ANDWF *f*,*d*

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(W) .AND. (f) \rightarrow (\text{destination})$

Status Affected: Z

Description: AND the W register with register 'f'. If 'd' is '0', the result is stored in the W register. If 'd' is '1', the result is stored back in register 'f'.

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BTFSS **Bit Test f, Skip if Set**

Syntax: [*label*] BTFSS *f*,*b*
Operands: $0 \leq f \leq 127$
 $0 \leq b < 7$
Operation: skip if (*f*<*b*>) = 1
Status Affected: None
Description: If bit '*b*' in register '*f*' is '0', the next instruction is executed.
 If bit '*b*' is '1', then the next instruction is discarded and a NOP is executed instead, making this a two-cycle instruction.

CLRWDT **Clear Watchdog Timer**

Syntax: [*label*] CLRWDT
Operands: None
Operation: 00h → WDT
 0 → WDT prescaler,
 1 → $\overline{TO}$
 1 → $\overline{PD}$
Status Affected: $\overline{TO}$, $\overline{PD}$
Description: CLRWDT instruction resets the Watchdog Timer. It also resets the prescaler of the WDT.
 Status bits $\overline{TO}$ and $\overline{PD}$ are set.

CALL **Call Subroutine**

Syntax: [*label*] CALL *k*
Operands: $0 \leq k \leq 2047$
Operation: (PC)+1 → TOS,
 k → PC<10:0>,
 (PCLATH<4:3>) → PC<12:11>
Status Affected: None
Description: Call Subroutine. First, return address (PC + 1) is pushed onto the stack. The eleven-bit immediate address is loaded into PC bits <10:0>. The upper bits of the PC are loaded from PCLATH. CALL is a two-cycle instruction.

COMF **Complement f**

Syntax: [*label*] COMF *f*,*d*
Operands: $0 \leq f \leq 127$
 d ∈ [0,1]
Operation: ($\bar{f}$) → (destination)
Status Affected: Z
Description: The contents of register '*f*' are complemented. If '*d*' is '0', the result is stored in W. If '*d*' is '1', the result is stored back in register '*f*'.

CLRF **Clear f**

Syntax: [*label*] CLRF *f*
Operands: $0 \leq f \leq 127$
Operation: 00h → (*f*)
 1 → Z
Status Affected: Z
Description: The contents of register '*f*' are cleared and the Z bit is set.

DECF **Decrement f**

Syntax: [*label*] DECF *f*,*d*
Operands: $0 \leq f \leq 127$
 d ∈ [0,1]
Operation: (*f*) - 1 → (destination)
Status Affected: Z
Description: Decrement register '*f*'. If '*d*' is '0', the result is stored in the W register. If '*d*' is '1', the result is stored back in register '*f*'.

CLRW **Clear W**

Syntax: [*label*] CLRW
Operands: None
Operation: 00h → (W)
 1 → Z
Status Affected: Z
Description: W register is cleared. Zero bit (Z) is set.

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DECFSZ Decrement f, Skip if 0

Syntax: **[label] DECFSZ f,d**

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(f) - 1 \rightarrow (\text{destination});$
skip if result = 0

Status Affected: None

Description: The contents of register 'f' are decremented. If 'd' is '0', the result is placed in the W register. If 'd' is '1', the result is placed back in register 'f'.
If the result is '1', the next instruction is executed. If the result is '0', then a NOP is executed instead, making it a two-cycle instruction.

INCFSZ Increment f, Skip if 0

Syntax: **[label] INCFSZ f,d**

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(f) + 1 \rightarrow (\text{destination}),$
skip if result = 0

Status Affected: None

Description: The contents of register 'f' are incremented. If 'd' is '0', the result is placed in the W register. If 'd' is '1', the result is placed back in register 'f'.
If the result is '1', the next instruction is executed. If the result is '0', a NOP is executed instead, making it a two-cycle instruction.

GOTO Unconditional Branch

Syntax: **[label] GOTO k**

Operands: $0 \leq k \leq 2047$

Operation: $k \rightarrow \text{PC}<10:0>$
 $\text{PCLATH}<4:3> \rightarrow \text{PC}<12:11>$

Status Affected: None

Description: GOTO is an unconditional branch. The eleven-bit immediate value is loaded into PC bits <10:0>. The upper bits of PC are loaded from PCLATH<4:3>. GOTO is a two-cycle instruction.

IORLW Inclusive OR literal with W

Syntax: **[label] IORLW k**

Operands: $0 \leq k \leq 255$

Operation: $(W) .OR. k \rightarrow (W)$

Status Affected: Z

Description: The contents of the W register are OR'ed with the eight-bit literal 'k'. The result is placed in the W register.

INCF Increment f

Syntax: **[label] INCF f,d**

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(f) + 1 \rightarrow (\text{destination})$

Status Affected: Z

Description: The contents of register 'f' are incremented. If 'd' is '0', the result is placed in the W register. If 'd' is '1', the result is placed back in register 'f'.

IORWF Inclusive OR W with f

Syntax: **[label] IORWF f,d**

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(W) .OR. (f) \rightarrow (\text{destination})$

Status Affected: Z

Description: Inclusive OR the W register with register 'f'. If 'd' is '0', the result is placed in the W register. If 'd' is '1', the result is placed back in register 'f'.

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MOVF	Move f
Syntax:	[<i>label</i>] MOVF f,d
Operands:	$0 \leq f \leq 127$ $d \in [0,1]$
Operation:	$(f) \rightarrow (\text{dest})$
Status Affected:	Z
Description:	The contents of register 'f' is moved to a destination dependent upon the status of 'd'. If $d = 0$, destination is W register. If $d = 1$, the destination is file register 'f' itself. $d = 1$ is useful to test a file register since status flag Z is affected.
Words:	1
Cycles:	1
Example:	MOVF FSR, 0 After Instruction W = value in FSR register Z = 1

MOVLW	Move literal to W
Syntax:	[<i>label</i>] MOVLW k
Operands:	$0 \leq k \leq 255$
Operation:	$k \rightarrow (W)$
Status Affected:	None
Description:	The eight-bit literal 'k' is loaded into W register. The "don't cares" will assemble as '0's.
Words:	1
Cycles:	1
Example:	MOVLW 0x5A After Instruction W = 0x5A

MOVWF	Move W to f
Syntax:	[<i>label</i>] MOVWF f
Operands:	$0 \leq f \leq 127$
Operation:	$(W) \rightarrow (f)$
Status Affected:	None
Description:	Move data from W register to register 'f'.
Words:	1
Cycles:	1
Example:	MOVW OPTION F Before Instruction OPTION = 0xFF W = 0x4F After Instruction OPTION = 0x4F W = 0x4F

NOP	No Operation
Syntax:	[<i>label</i>] NOP
Operands:	None
Operation:	No operation
Status Affected:	None
Description:	No operation.
Words:	1
Cycles:	1
Example:	NOP

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RETFIE	Return from Interrupt
Syntax:	[<i>label</i>] RETFIE
Operands:	None
Operation:	TOS → PC, 1 → GIE
Status Affected:	None
Description:	Return from Interrupt. Stack is POPed and Top-of-Stack (TOS) is loaded in the PC. Interrupts are enabled by setting Global Interrupt Enable bit, GIE (INTCON<7>). This is a two-cycle instruction.
Words:	1
Cycles:	2
Example:	RETFIE After Interrupt PC = TOS GIE = 1

RETLW	Return with literal in W
Syntax:	[<i>label</i>] RETLW k
Operands:	0 ≤ k ≤ 255
Operation:	k → (W); TOS → PC
Status Affected:	None
Description:	The W register is loaded with the eight-bit literal 'k'. The program counter is loaded from the top of the stack (the return address). This is a two-cycle instruction.
Words:	1
Cycles:	2
Example:	CALL TABLE;W contains table ;offset value TABLE • ;W now has • ;table value • • ADDWF PC ;W = offset RETLW k1 ;Begin table RETLW k2 ; • • • RETLW kn ;End of table Before Instruction W = 0x07 After Instruction W = value of k8

RETURN	Return from Subroutine
Syntax:	[<i>label</i>] RETURN
Operands:	None
Operation:	TOS → PC
Status Affected:	None
Description:	Return from subroutine. The stack is POPed and the top of the stack (TOS) is loaded into the program counter. This is a two-cycle instruction.

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RLF Rotate Left f through Carry

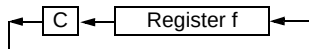
Syntax: [label] RLF f,d

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: See description below

Status Affected: C

Description: The contents of register 'f' are rotated one bit to the left through the Carry flag. If 'd' is '0', the result is placed in the W register. If 'd' is '1', the result is stored back in register 'f'.



Words: 1

Cycles: 1

Example: RLF REG1, 0

Before Instruction

REG1 = 1110 0110
C = 0

After Instruction

REG1 = 1110 0110
W = 1100 1100
C = 1

SLEEP Enter Sleep mode

Syntax: [label] SLEEP

Operands: None

Operation: 00h → WDT,
0 → WDT prescaler,
1 → $\overline{TO}$,
0 → PD

Status Affected: $\overline{TO}$, PD

Description: The power-down Status bit, $\overline{PD}$ is cleared. Time-out Status bit, $\overline{TO}$ is set. Watchdog Timer and its prescaler are cleared. The processor is put into Sleep mode with the oscillator stopped.

RRF Rotate Right f through Carry

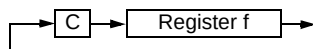
Syntax: [label] RRF f,d

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: See description below

Status Affected: C

Description: The contents of register 'f' are rotated one bit to the right through the Carry flag. If 'd' is '0', the result is placed in the W register. If 'd' is '1', the result is placed back in register 'f'.



SUBLW Subtract W from literal

Syntax: [label] SUBLW k

Operands: $0 \leq k \leq 255$

Operation: $k - (W) \rightarrow (W)$

Status Affected: C, DC, Z

Description: The W register is subtracted (2's complement method) from the eight-bit literal 'k'. The result is placed in the W register.

C = 0	$W > k$
C = 1	$W \leq k$
DC = 0	$W<3:0> > k<3:0>$
DC = 1	$W<3:0> \leq k<3:0>$

SUBWF Subtract W from f

Syntax: `[label] SUBWF f,d`

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(f) - (W) \rightarrow (\text{destination})$

Status Affected: C, DC, Z

Description: Subtract (2's complement method) W register from register 'f'. If 'd' is '0', the result is stored in the W register. If 'd' is '1', the result is stored back in register 'f'.

C = 0	$W > f$
C = 1	$W \leq f$
DC = 0	$W<3:0> > f<3:0>$
DC = 1	$W<3:0> \leq f<3:0>$

XORWF Exclusive OR W with f

Syntax: `[label] XORWF f,d`

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(W) .XOR. (f) \rightarrow (\text{destination})$

Status Affected: Z

Description: Exclusive OR the contents of the W register with register 'f'. If 'd' is '0', the result is stored in the W register. If 'd' is '1', the result is stored back in register 'f'.

SWAPF Swap Nibbles in f

Syntax: `[label] SWAPF f,d`

Operands: $0 \leq f \leq 127$
 $d \in [0,1]$

Operation: $(f<3:0>) \rightarrow (\text{destination}<7:4>),$
 $(f<7:4>) \rightarrow (\text{destination}<3:0>)$

Status Affected: None

Description: The upper and lower nibbles of register 'f' are exchanged. If 'd' is '0', the result is placed in the W register. If 'd' is '1', the result is placed in register 'f'.

XORLW Exclusive OR literal with W

Syntax: `[label] XORLW k`

Operands: $0 \leq k \leq 255$

Operation: $(W) .XOR. k \rightarrow (W)$

Status Affected: Z

Description: The contents of the W register are XOR'ed with the eight-bit literal 'k'. The result is placed in the W register.

17.0 ELECTRICAL SPECIFICATIONS

Absolute Maximum Ratings^(†)

Ambient temperature under bias	-40° to +125°C
Storage temperature	-65°C to +150°C
Voltage on V _{DD} with respect to V _{SS}	-0.3V to +6.5V
Voltage on $\overline{\text{MCLR}}$ with respect to V _{SS}	-0.3V to +13.5V
Voltage on all other pins with respect to V _{SS}	-0.3V to (V _{DD} + 0.3V)
Total power dissipation ⁽¹⁾	800 mW
Maximum current out of V _{SS} pin	95 mA
Maximum current into V _{DD} pin	95 mA
Input clamp current, I _{IK} (V _I < 0 or V _I > V _{DD})	± 20 mA
Output clamp current, I _{OK} (V _O < 0 or V _O > V _{DD})	± 20 mA
Maximum output current sunk by any I/O pin	25 mA
Maximum output current sourced by any I/O pin	25 mA
Maximum current sunk by all ports (combined) ⁽²⁾	90 mA
Maximum current sourced by all ports (combined) ⁽²⁾	90 mA

Note 1: Power dissipation is calculated as follows: $P_{DIS} = V_{DD} \times \{I_{DD} - \sum I_{OH}\} + \sum \{(V_{DD} - V_{OH}) \times I_{OH}\} + \sum (V_{OL} \times I_{OL})$.

2: PORTD and PORTE are implemented on PIC16F886/PIC16F887 only.

† NOTICE: Stresses above those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. This is a stress rating only and functional operation of the device at those or any other conditions above those indicated in the operation listings of this specification is not implied. Exposure above maximum rating conditions for extended periods may affect device reliability.